



EUROPEAN
SPALLATION
SOURCE



Same workflow, different techniques

From a diffraction prototype to a reusable EasyScience GUI

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NOBUGS 2026 · GUI Workshop · 21 Sep 2026

- 1 🔍 THE PROBLEM
- 2 🛠️ DESIGN EVOLUTION
- 3 🧠 UX PRINCIPLES
- 4 📦 REUSE
- 5 🖥️ DEMO

1 🔍 THE PROBLEM

2 🛠️ DESIGN EVOLUTION

3 🧠 UX PRINCIPLES

4 🧱 REUSE

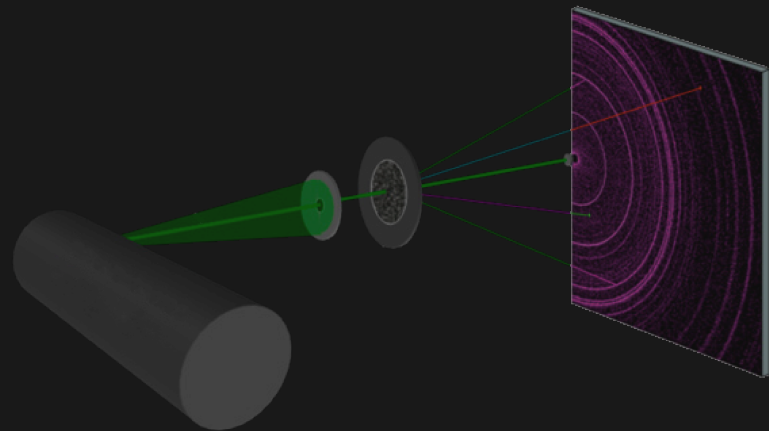
5 🖥️ DEMO

Where analysis sits

Analysis is where each technique has *its own tools*.

Acquisition

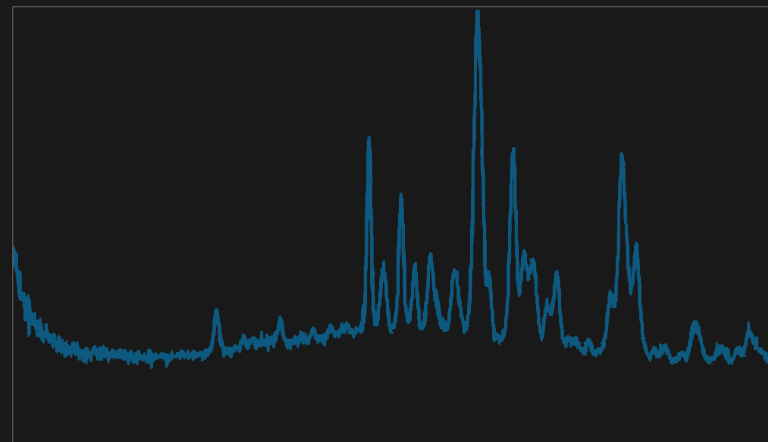
run the experiment, write raw events



NICOS

Reduction

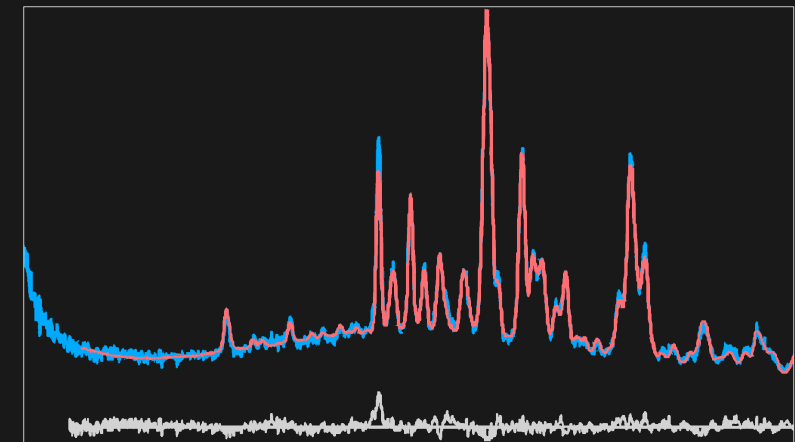
correct, bin, turn events into a pattern



scipp

Analysis

fit a model, extract the science



easy
diffraction

SasView

+ many technique-specific tools

1.2 🔍 THE PROBLEM

One stage, many tools

Diffraction tools. Other techniques have their own.

DIFFRACTION PROGRAMS

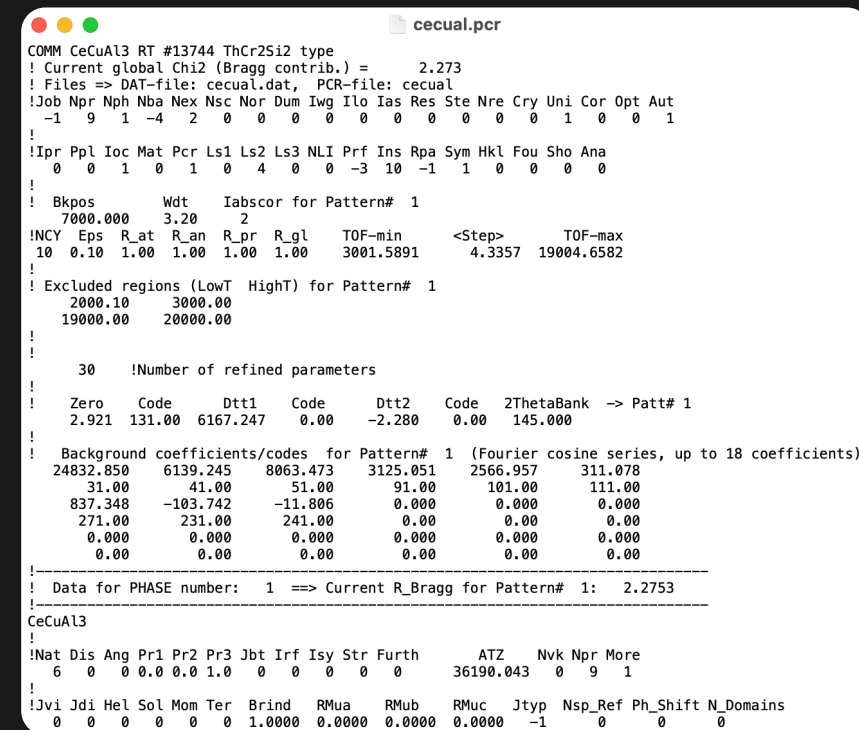
FullProf · Jana2020 · GSAS-II · SHELX · MAUD ·
FOX · Mag2Pol · TOPAS · HighScore · ...

DIFFRACTION LIBRARIES

CrysFML · CCSL · CCTBX · pyobjcryst · CrysPy ·
GSAS-II · ...

The science is excellent. Getting started is not.

Hard to start
platform-specific, extra software to install



OK

1.4 🔍 THE PROBLEM

The same work starts again

Different science. Same interface **rebuilt** every time.

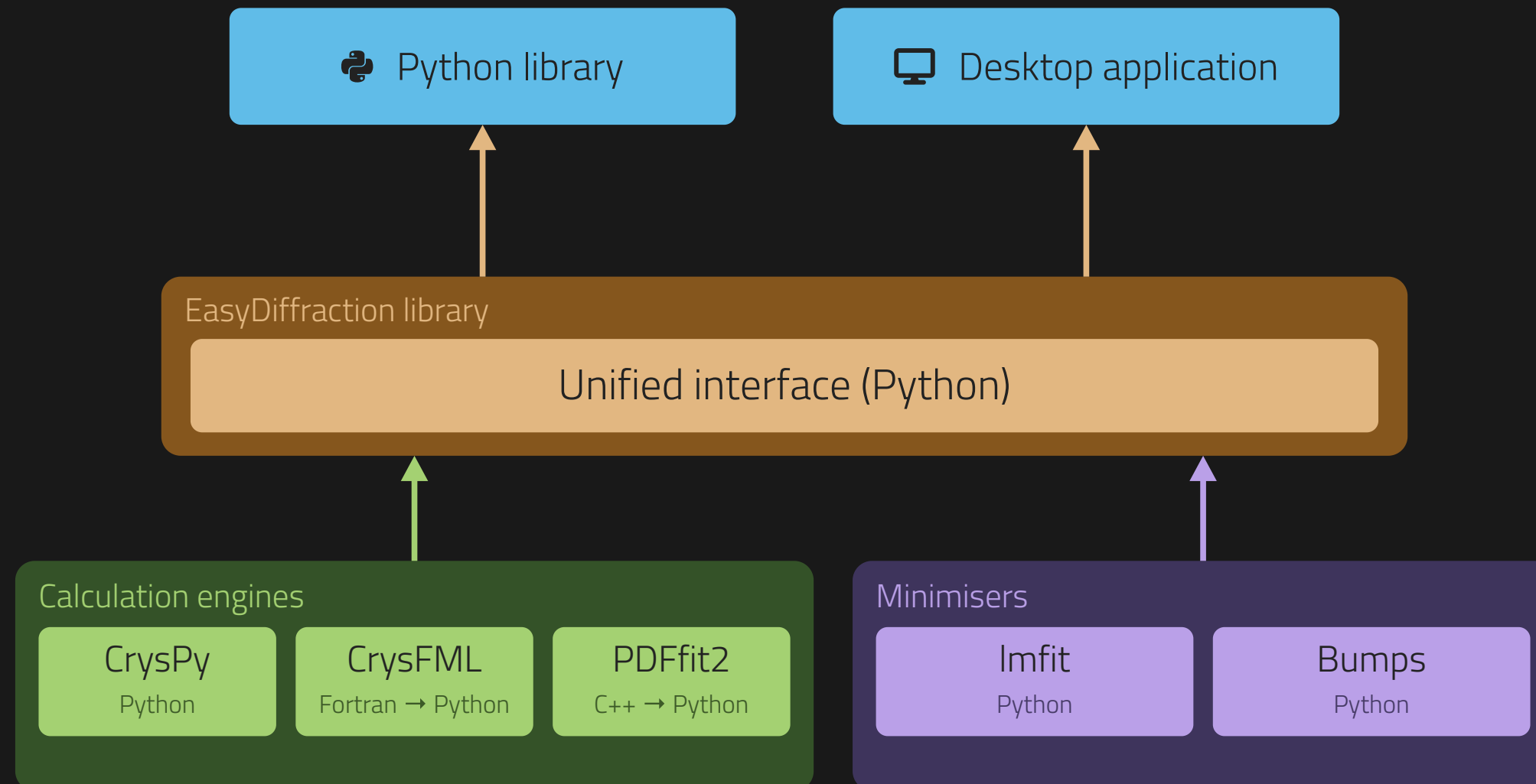
⚙️ A new technique gets a **new program**

💻 A new program gets a **new interface**

👤 Every usability lesson is learned once, then **lost**

What we build, what we reuse

We reused the calculations. We **built the interface**.



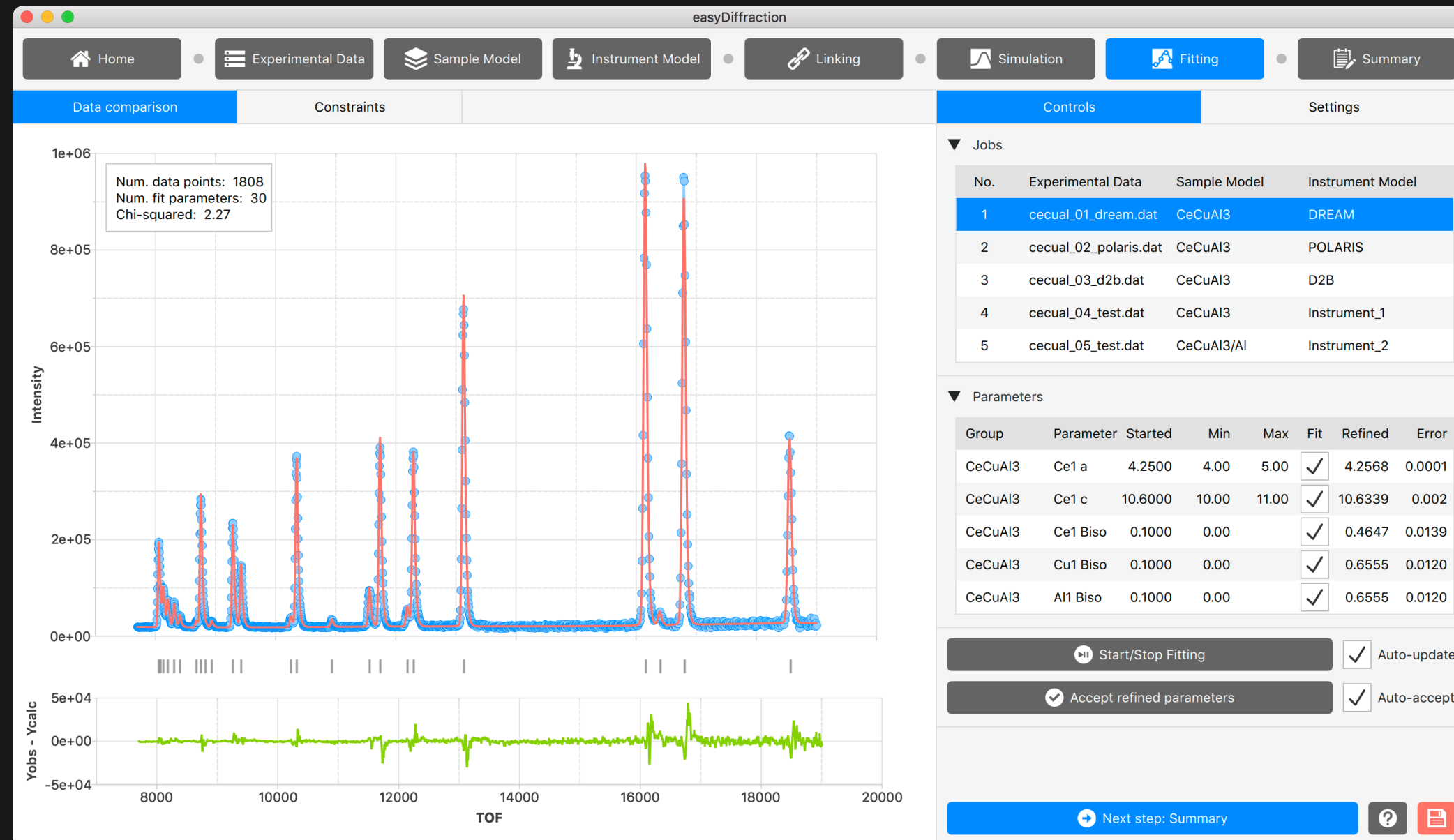
- 1 🔍 THE PROBLEM
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“Perfection is achieved, not when there is nothing more to add, but when there is **nothing left to take away.**”

Antoine de Saint-Exupéry

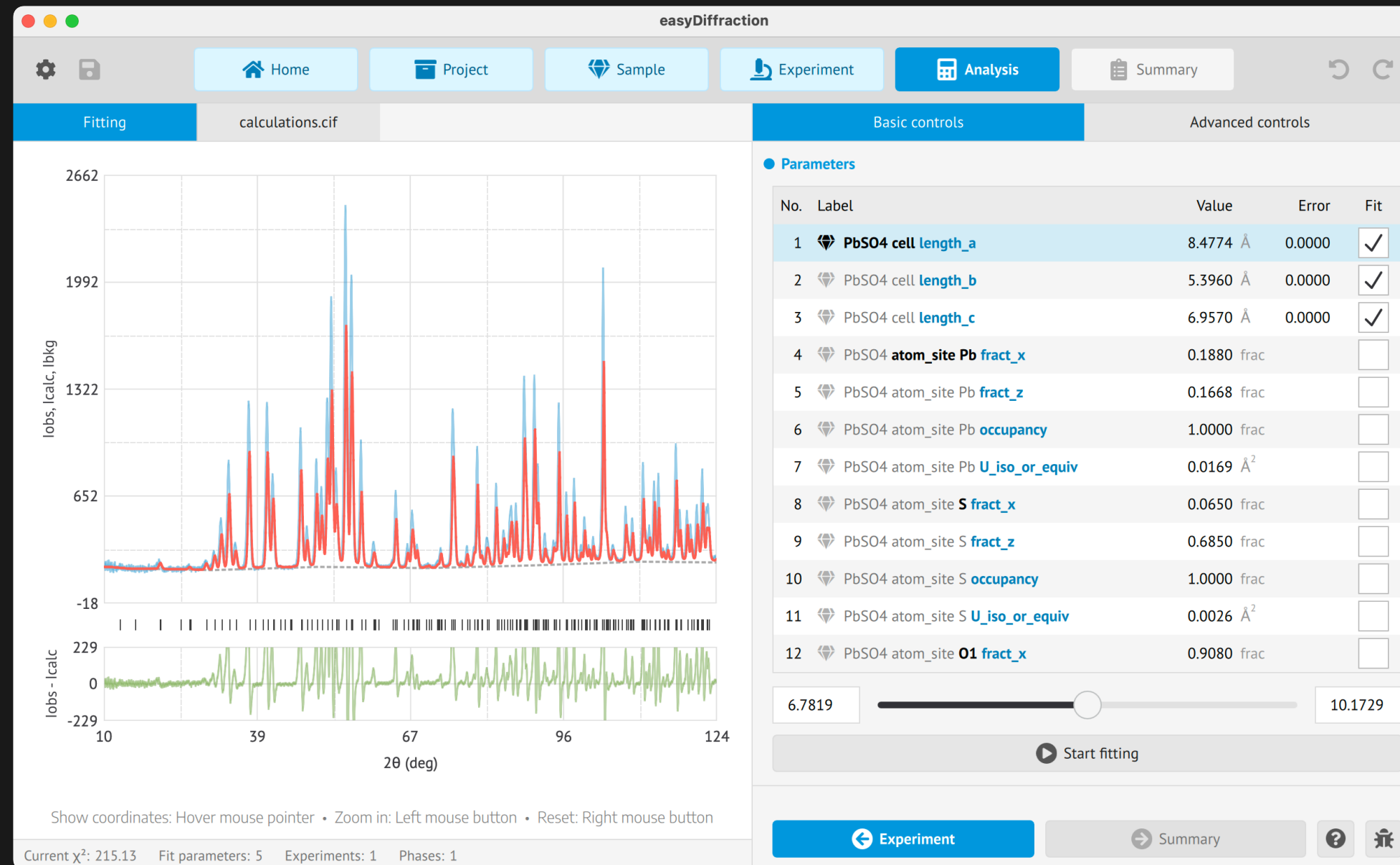
Several years of taking things away

2019 · the first mock-up, for diffraction



Several years of taking things away

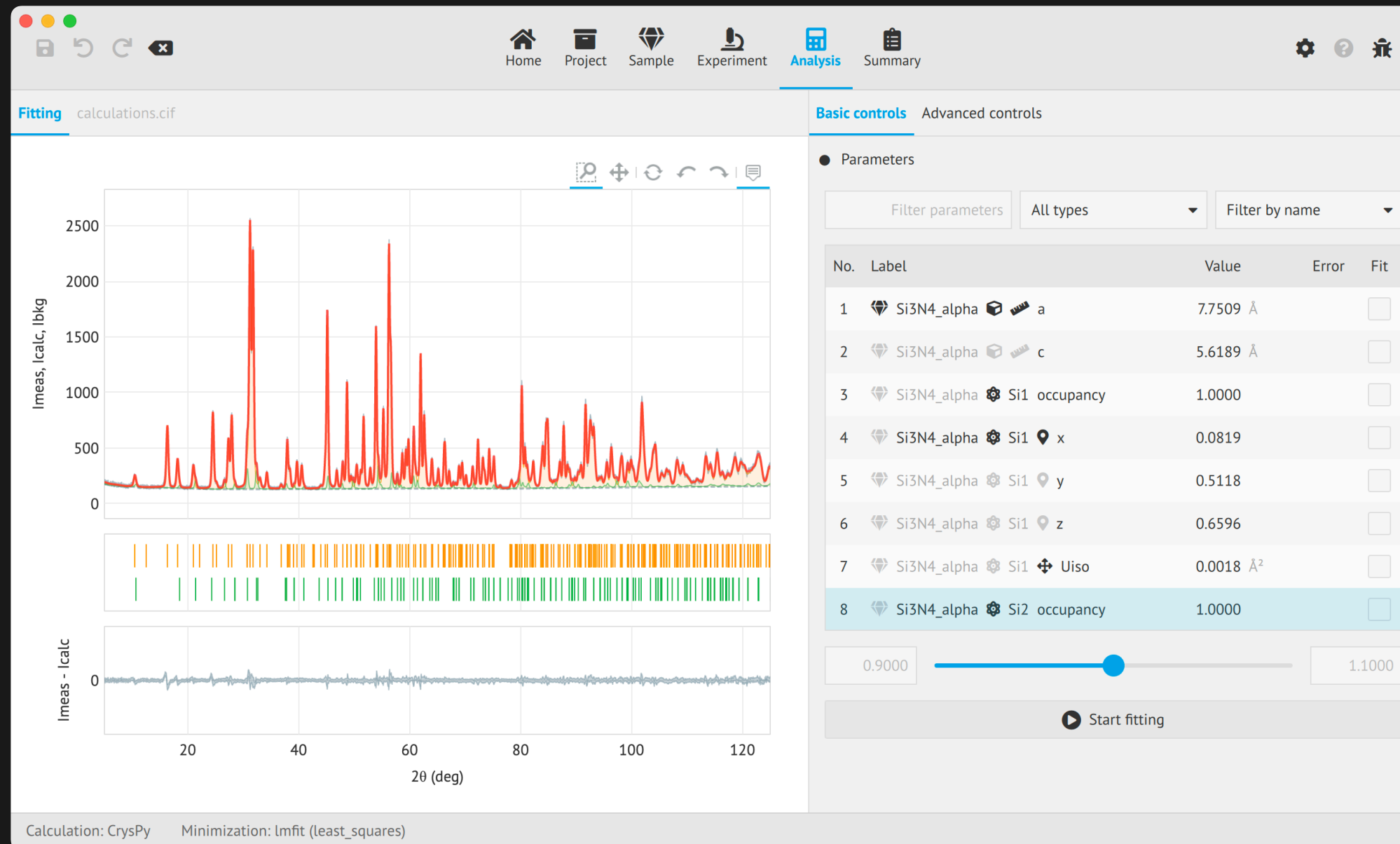
2020 · eight tabs became **six**



Several years of taking things away

10 | 26

2021 · simpler, cleaner, lighter



Several years of taking things away

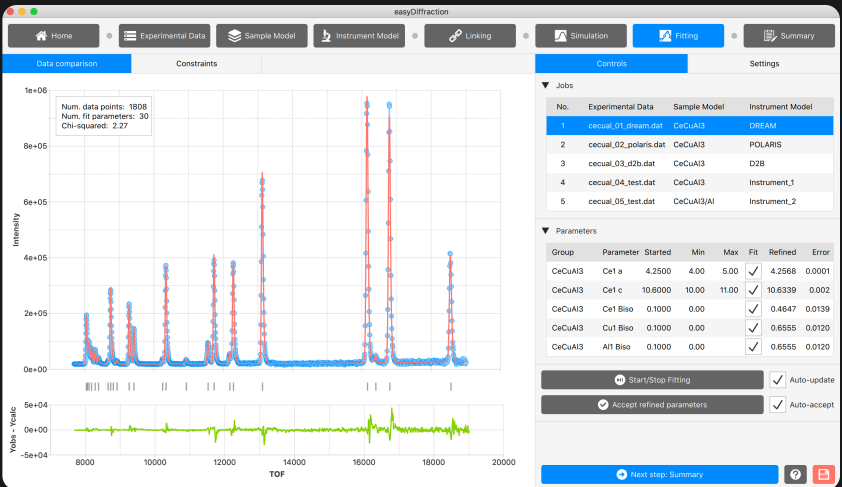
10 | 26

2023–today · the same shape, [easier to use](#)

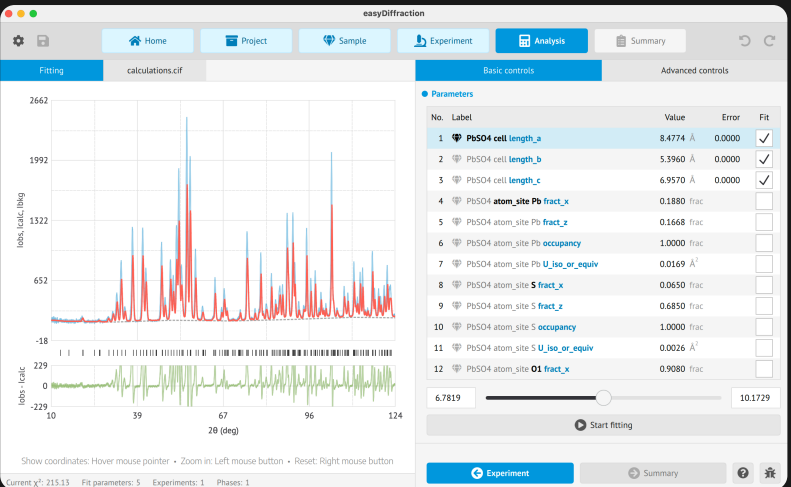
Several years of taking things away

Almost every iteration **removed** something.

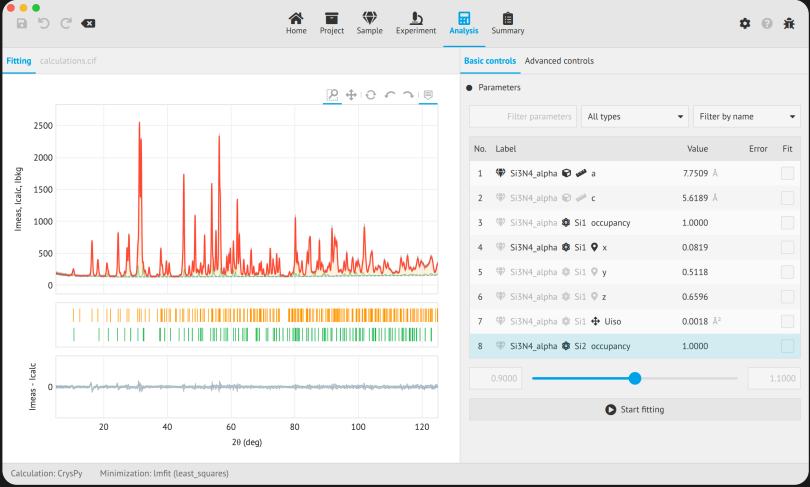
2019



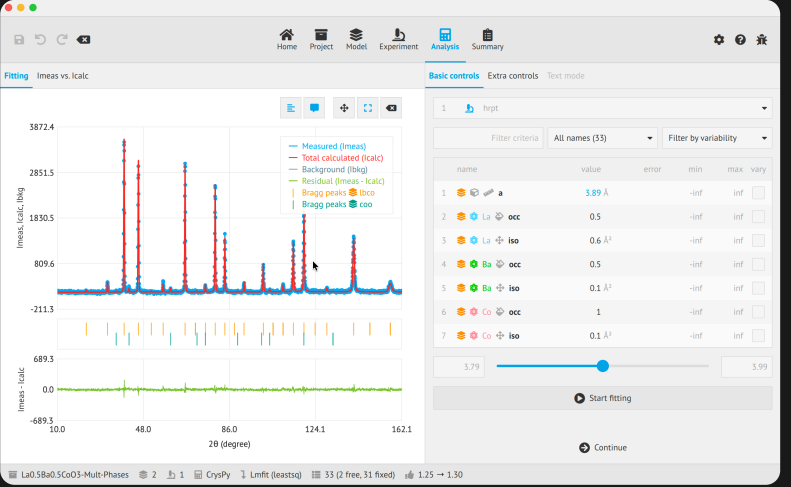
2020



2021

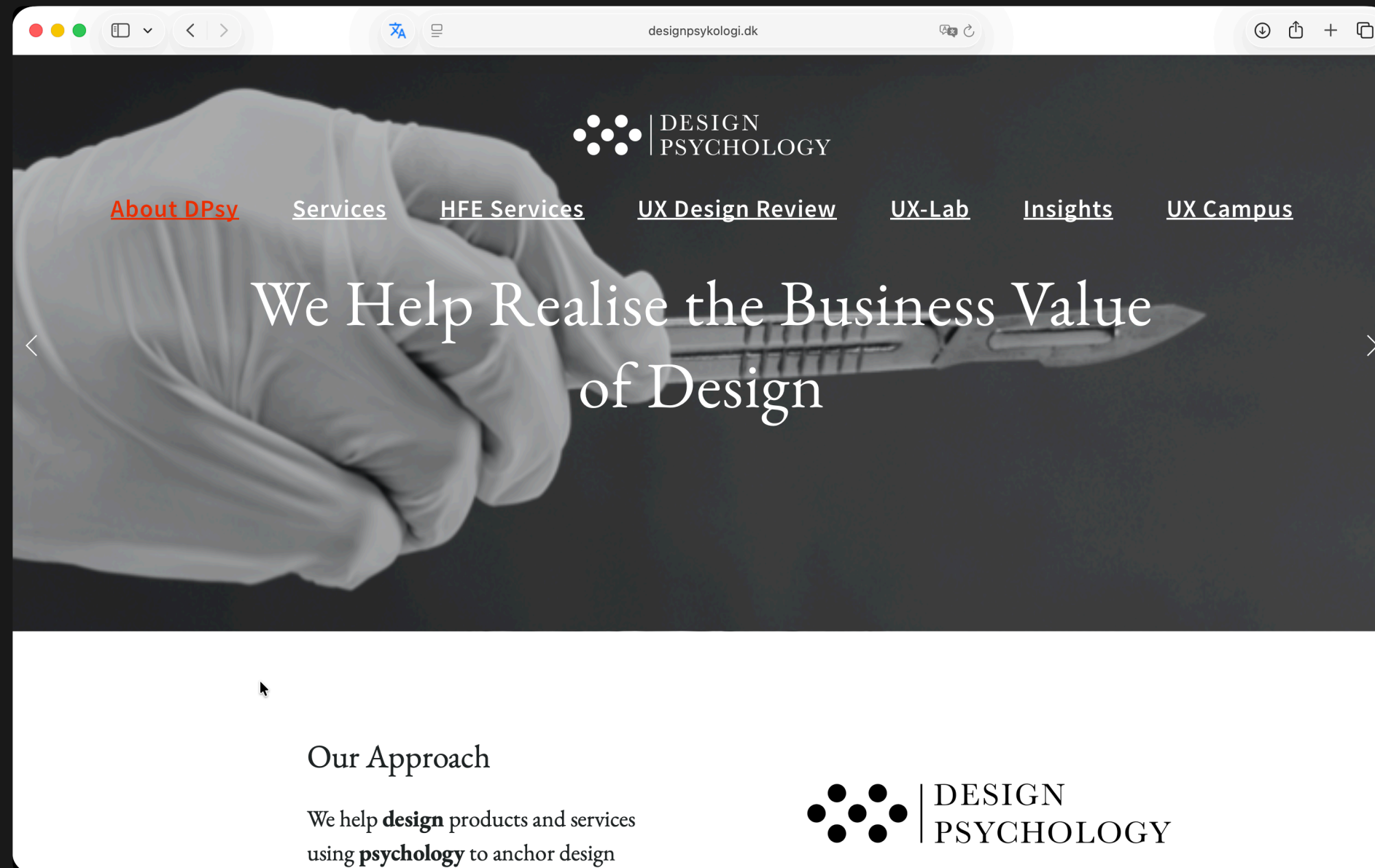


2023–today



Then we asked the experts

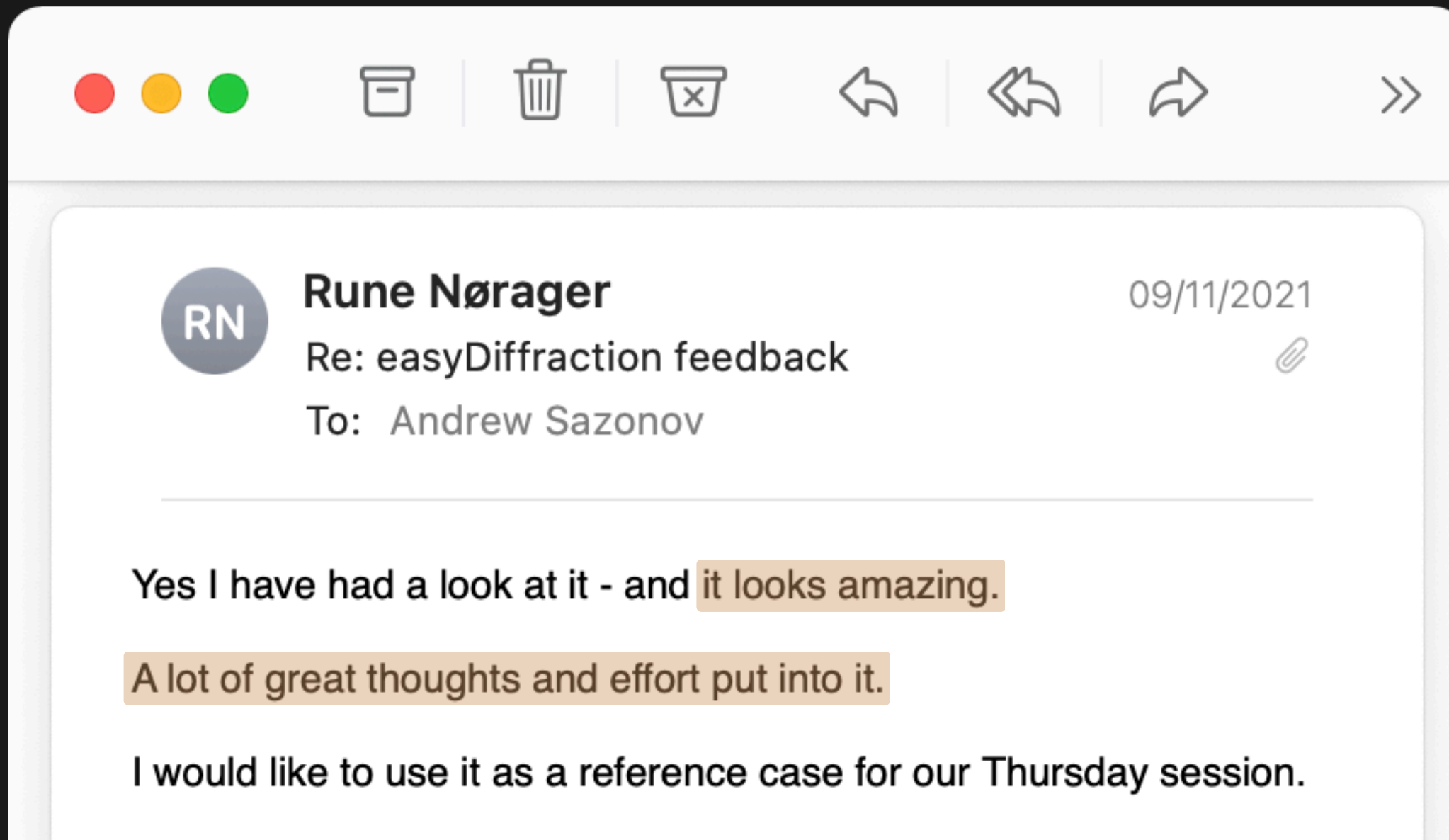
Two workshops with [Design Psychology](#), and a UX Expert Review.



Then we asked the experts

 **Approved** by the experts.

Then they reviewed **our interface**.



2.3 DESIGN EVOLUTION

The interface came before the science

Techniques that share the workflow **share the interface.**

THE USUAL ORDER

`</>` code →  interface

The interface takes its shape from the code.

OURS

 interface → `</>` code

The interface can be thrown away and redrawn.

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3.1 UX PRINCIPLES

Seeing beats reading

The same fact twice. Only one takes **effort**.

READ IT

"Analysis and Summary are not available until an experiment has been loaded."

Find it, understand it, remember it.

SEE IT



Home



Project



Model

**Experiment**

Analysis

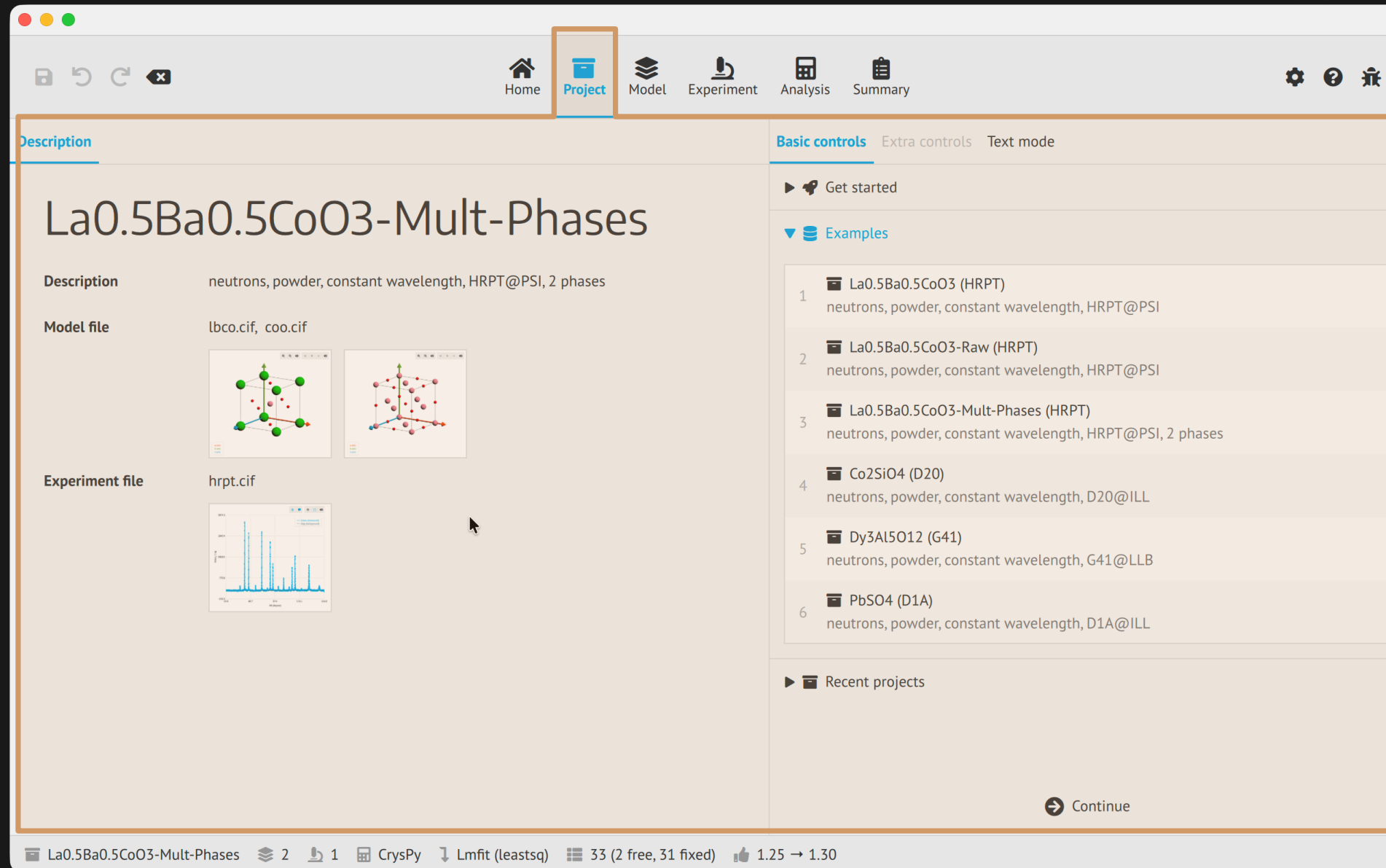


Summary

Two greyed-out words. Nothing to remember.

One path through the data

Name the [project](#), choose where to save it.



The screenshot shows a web-based application interface for managing scientific projects. The top navigation bar includes icons for Home, Project (highlighted with an orange box), Model, Experiment, Analysis, and Summary. The main content area is divided into two panels. The left panel, titled 'Description', displays the project name 'La0.5Ba0.5CoO3-Mult-Phases' and provides details about the experiment (neutrons, powder, constant wavelength, HRPT@PSI, 2 phases), model files (lbco.cif, coo.cif), and experiment files (hrpt.cif). It includes visualizations of crystal structures and a neutron diffraction pattern. The right panel, titled 'Basic controls', offers tabs for 'Basic controls', 'Extra controls', and 'Text mode'. It features a 'Get started' button, a list of 'Examples' (numbered 1 to 6) with project names and descriptions, and a 'Recent projects' section. A 'Continue' button is located at the bottom right of the right panel. The bottom status bar shows the current project name, a list of installed packages (Cryspy, Lmfit (leastsq)), and a progress indicator (33 (2 free, 31 fixed) and 1.25 → 1.30).

La0.5Ba0.5CoO3-Mult-Phases

Description: neutrons, powder, constant wavelength, HRPT@PSI, 2 phases

Model file: lbco.cif, coo.cif

Experiment file: hrpt.cif

Basic controls | Extra controls | Text mode

▶ Get started

▼ Examples

- 1 La0.5Ba0.5CoO3 (HRPT)
neutrons, powder, constant wavelength, HRPT@PSI
- 2 La0.5Ba0.5CoO3-Raw (HRPT)
neutrons, powder, constant wavelength, HRPT@PSI
- 3 La0.5Ba0.5CoO3-Mult-Phases (HRPT)
neutrons, powder, constant wavelength, HRPT@PSI, 2 phases
- 4 Co2SiO4 (D20)
neutrons, powder, constant wavelength, D20@ILL
- 5 Dy3Al5O12 (G41)
neutrons, powder, constant wavelength, G41@LLB
- 6 PbSO4 (D1A)
neutrons, powder, constant wavelength, D1A@ILL

▶ Recent projects

Continue

La0.5Ba0.5CoO3-Mult-Phases 2 1 Cryspy Lmfit (leastsq) 33 (2 free, 31 fixed) 1.25 → 1.30

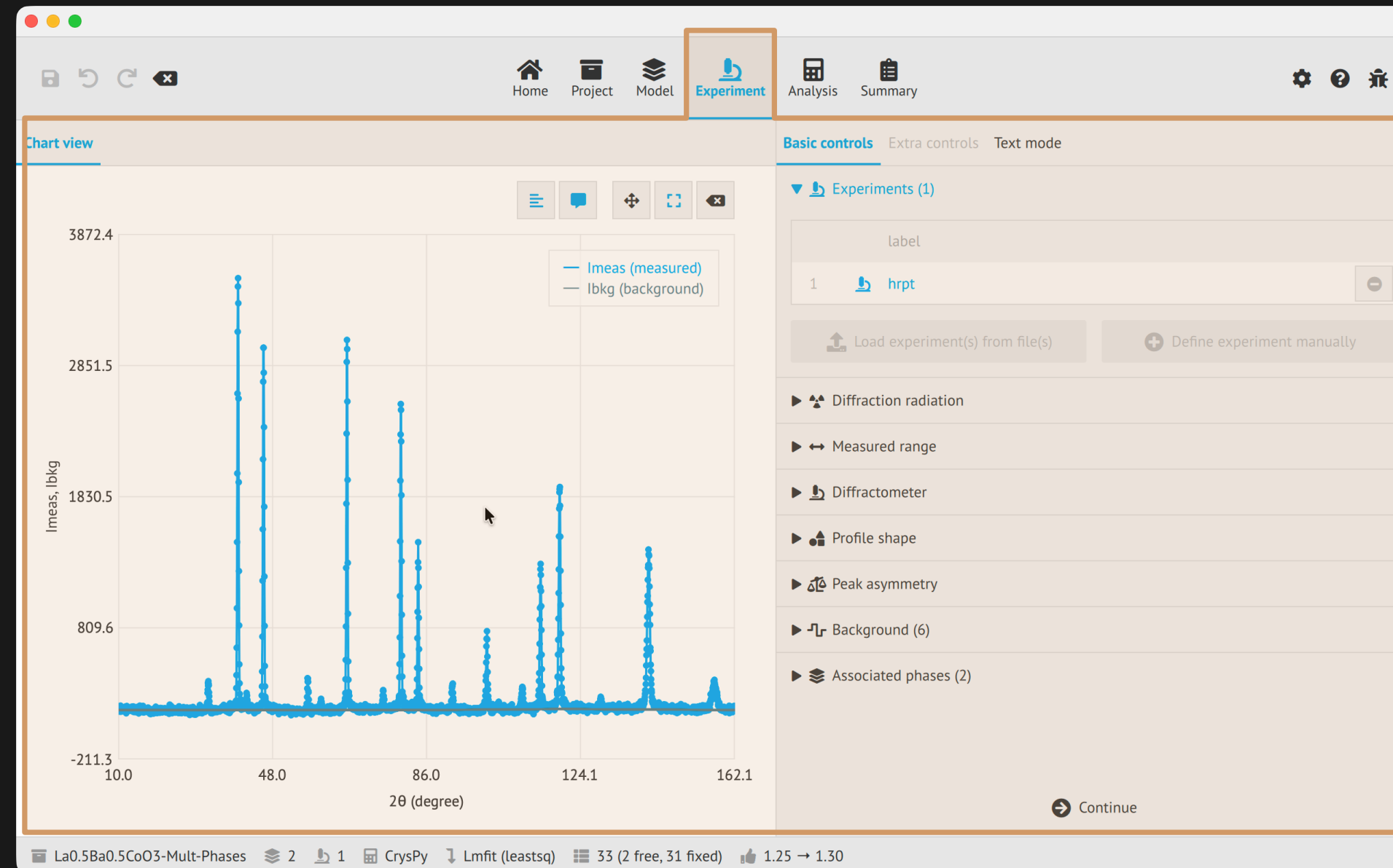
One path through the data

Describe the [model](#) — and see it.

The screenshot displays a software interface for crystal structure modeling. The main window is titled "Structure view" and shows a 3D model of a crystal structure. The model consists of green spheres (atoms) arranged in a cubic lattice, with red spheres (oxygen atoms) interspersed. A legend in the bottom-left corner identifies the axes: "a axis" (red), "b axis" (green), and "c axis" (blue). The interface includes a top navigation bar with icons for Home, Project, Model (highlighted), Experiment, Analysis, and Summary. A sidebar on the right contains "Basic controls" and "Extra controls" tabs. The "Basic controls" tab shows a list of models (1: lbco, 2: coo) and buttons for "Load model(s) from file(s)" and "Define model manually". Below this, there are expandable sections for "Space group", "Cell", "Atom site (4)", and "Atomic displacement (4)". A "Continue" button is located at the bottom right of the sidebar. The bottom status bar shows the chemical formula "La0.5Ba0.5CoO3-Mult-Phases", the number of models (2), the number of atoms (1), the software used (CrySPy), the fitting method (Lmfit (leastsq)), the number of parameters (33 (2 free, 31 fixed)), and the R-factor (1.25 → 1.30).

One path through the data

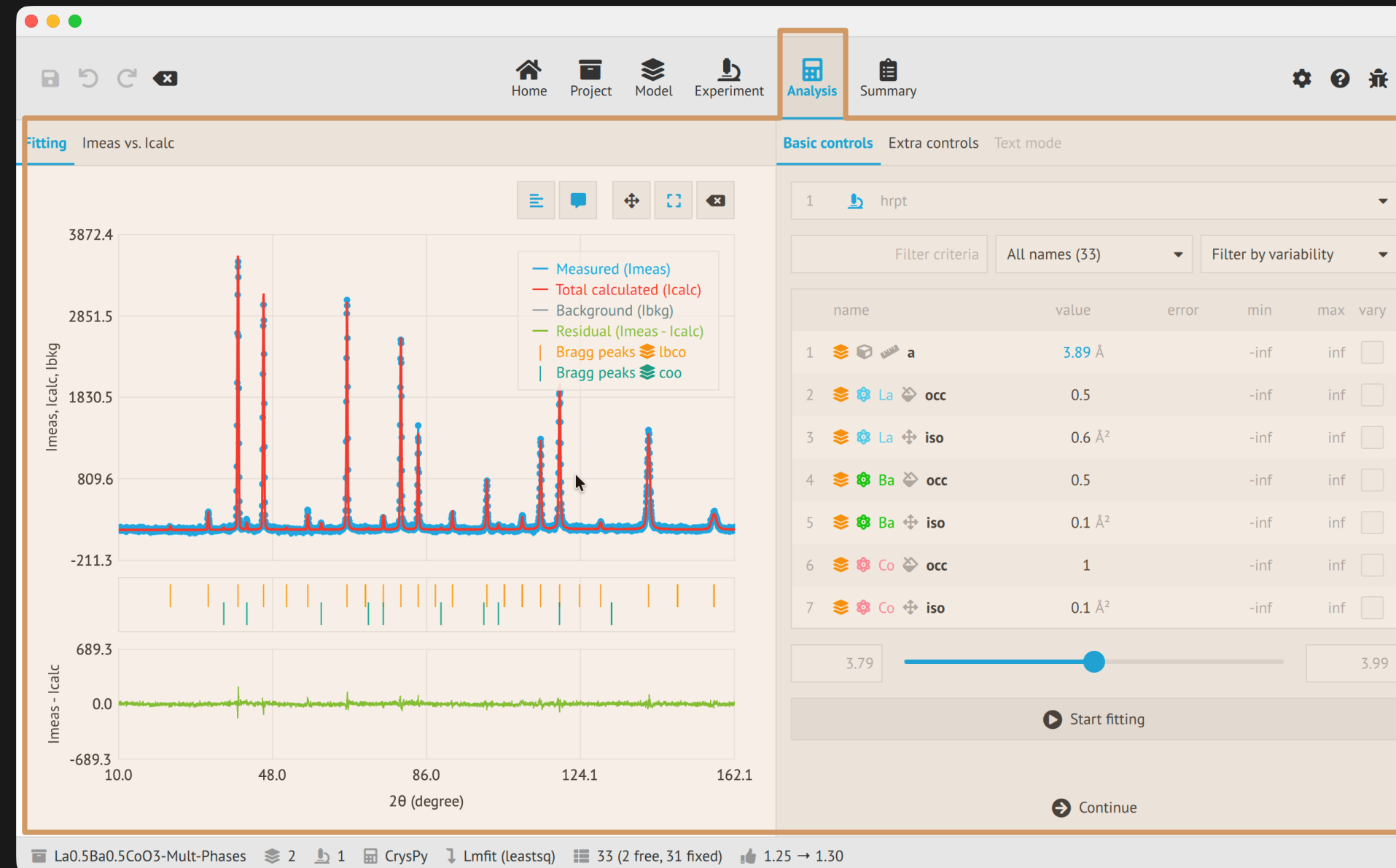
Load the [measurement](#) and its instrument.



One path through the data

15 | 26

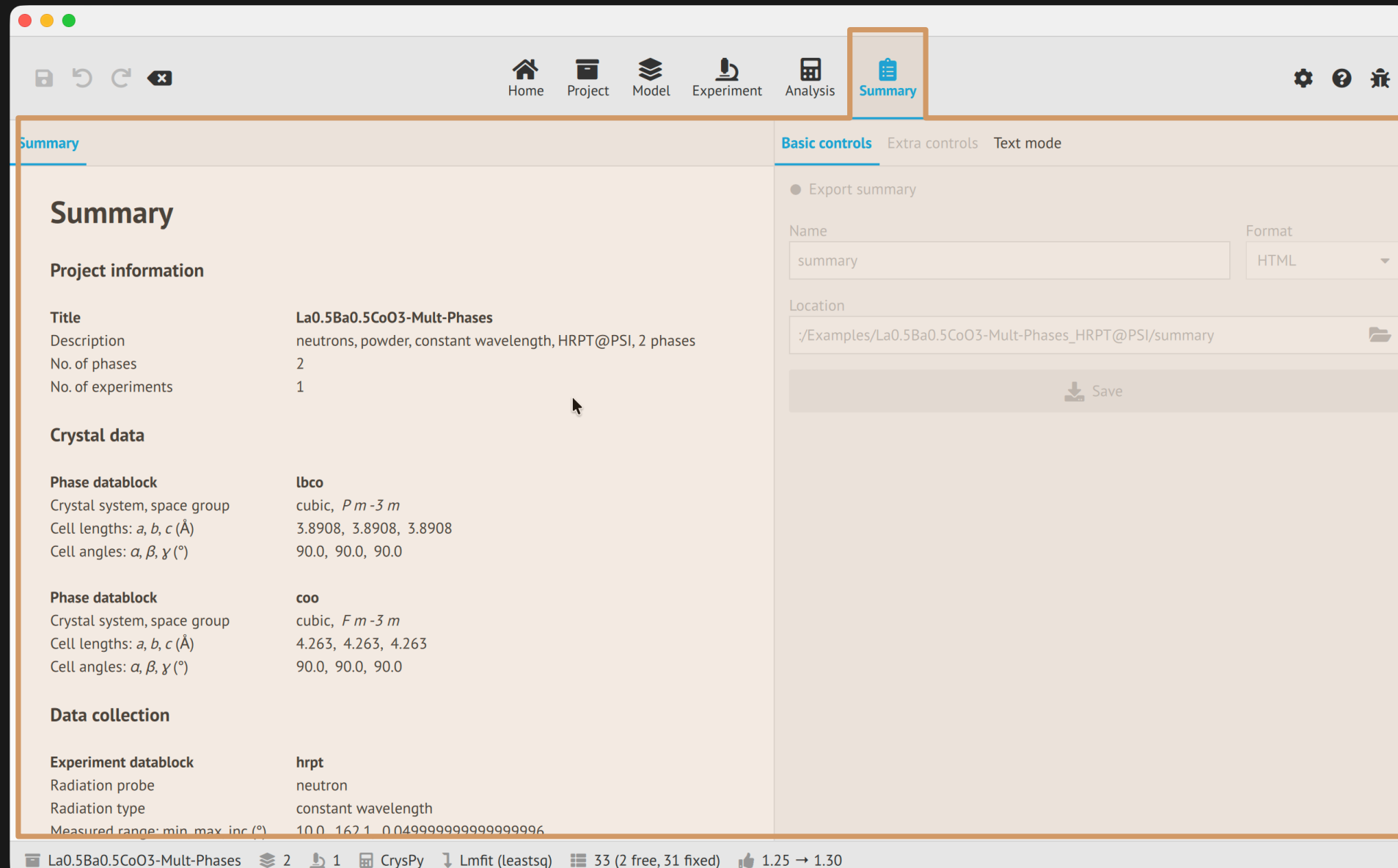
Choose *what to vary*, watch the fit run.



One path through the data

The tabs are the **steps of the analysis**.

Export a **report** using the same words as the screen.



The screenshot displays a software interface for data analysis. The top navigation bar includes icons for Home, Project, Model, Experiment, Analysis, and Summary (highlighted). The main content area is divided into two panels. The left panel, titled 'Summary', contains sections for Project information, Crystal data, and Data collection. The right panel, titled 'Basic controls', contains an 'Export summary' section with fields for Name, Format, and Location, and a 'Save' button. The bottom status bar shows the current project name, number of phases, and other details.

Project information	
Title	La0.5Ba0.5CoO3-Mult-Phases
Description	neutrons, powder, constant wavelength, HRPT@PSI, 2 phases
No. of phases	2
No. of experiments	1

Crystal data	
Phase datablock	
Crystal system, space group	lbco
Cell lengths: a, b, c (Å)	cubic, $Pm\bar{3}m$
Cell angles: α, β, γ (°)	3.8908, 3.8908, 3.8908
Phase datablock	
Crystal system, space group	coo
Cell lengths: a, b, c (Å)	cubic, $Fm\bar{3}m$
Cell angles: α, β, γ (°)	4.263, 4.263, 4.263

Data collection	
Experiment datablock	
Radiation probe	hrpt
Radiation type	neutron
Measured range: min_max_inc (°)	constant wavelength
10.0 162.1 0.049999999999999996	

Export summary

Name: summary Format: HTML

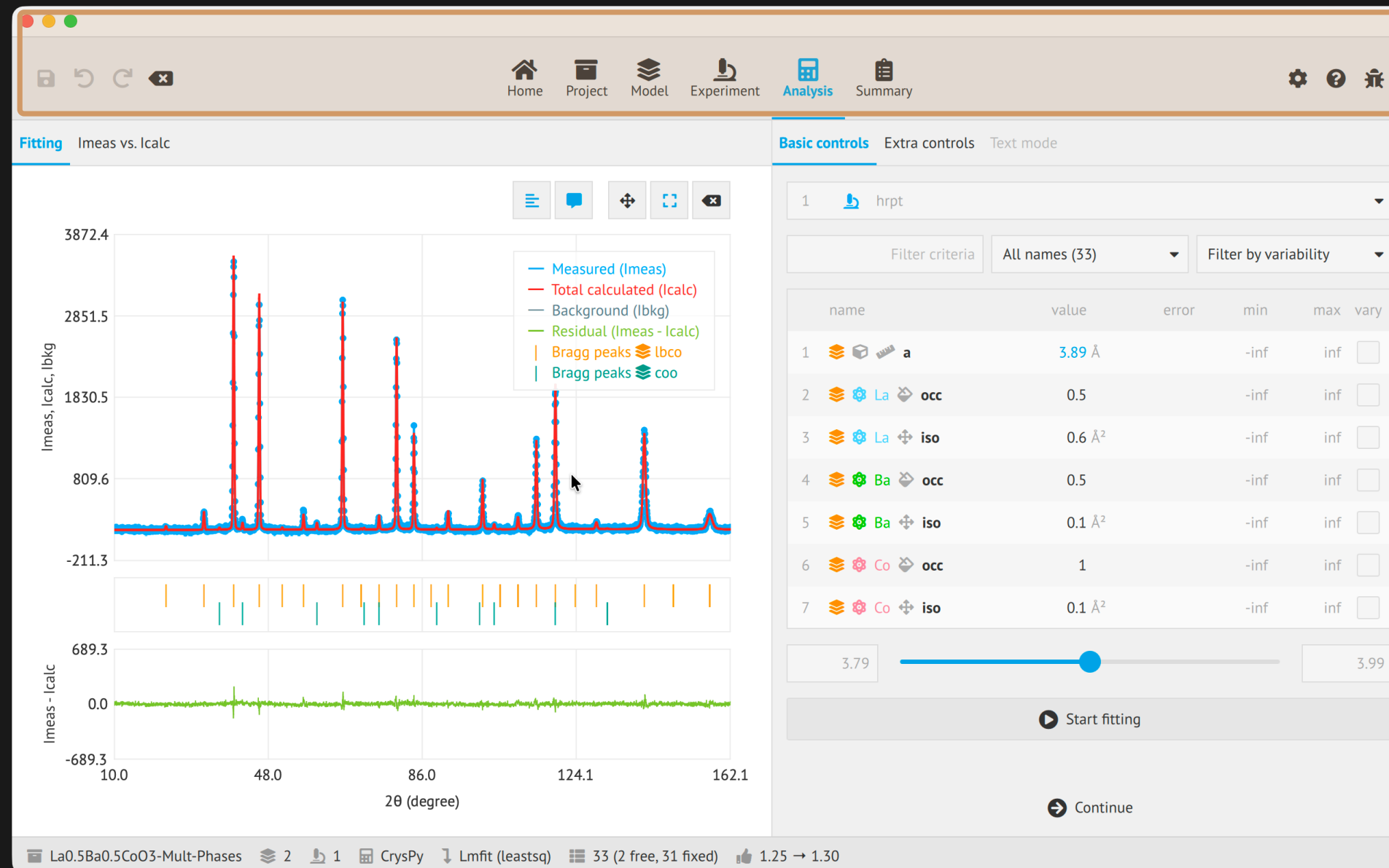
Location: /Examples/La0.5Ba0.5CoO3-Mult-Phases_HRPT@PSI/summary

Save

La0.5Ba0.5CoO3-Mult-Phases 2 1 CrysPy Lmfit (leastsq) 33 (2 free, 31 fixed) 1.25 → 1.30

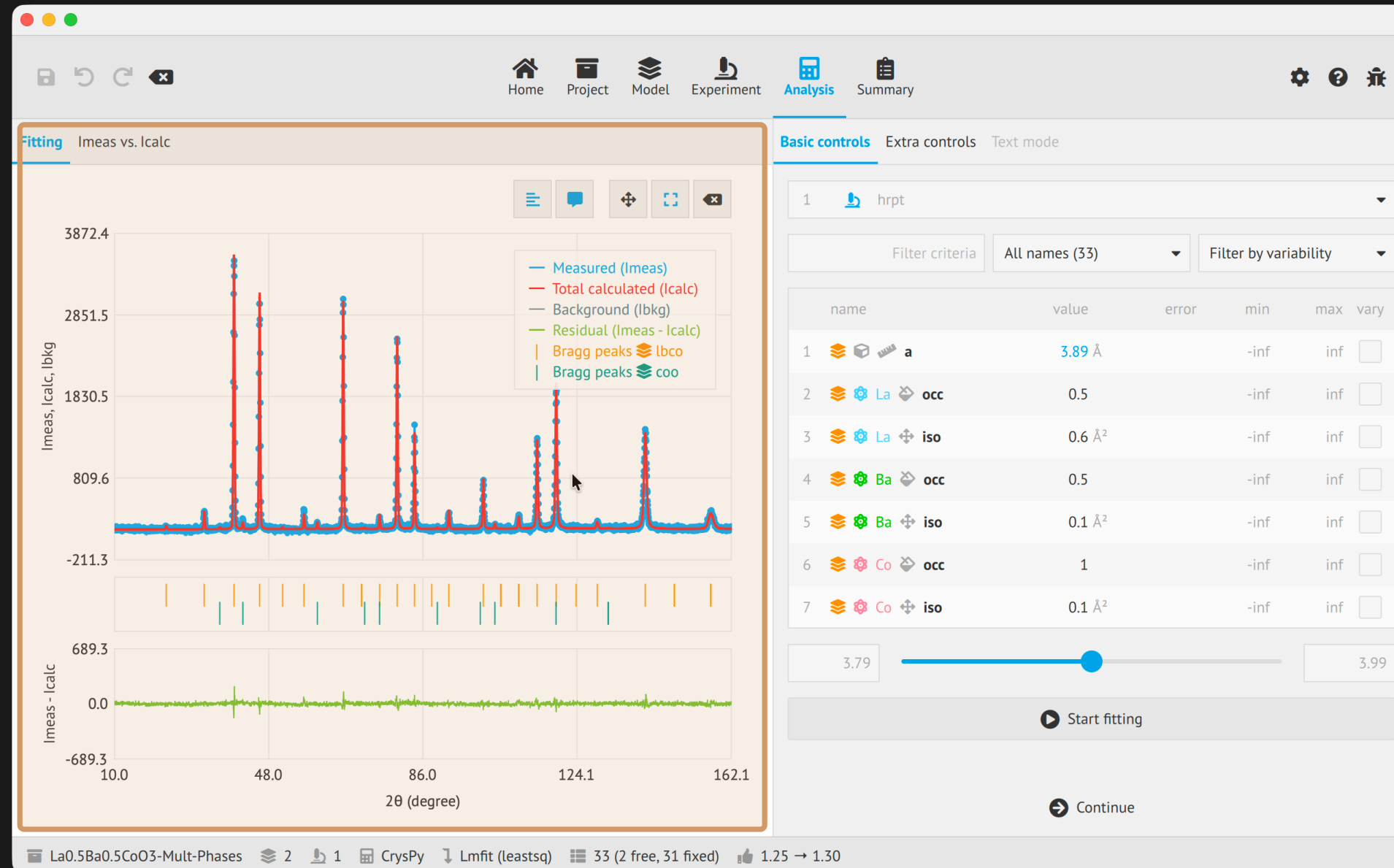
One screen, four zones

Where you are — the same tab bar on every screen.



One screen, four zones

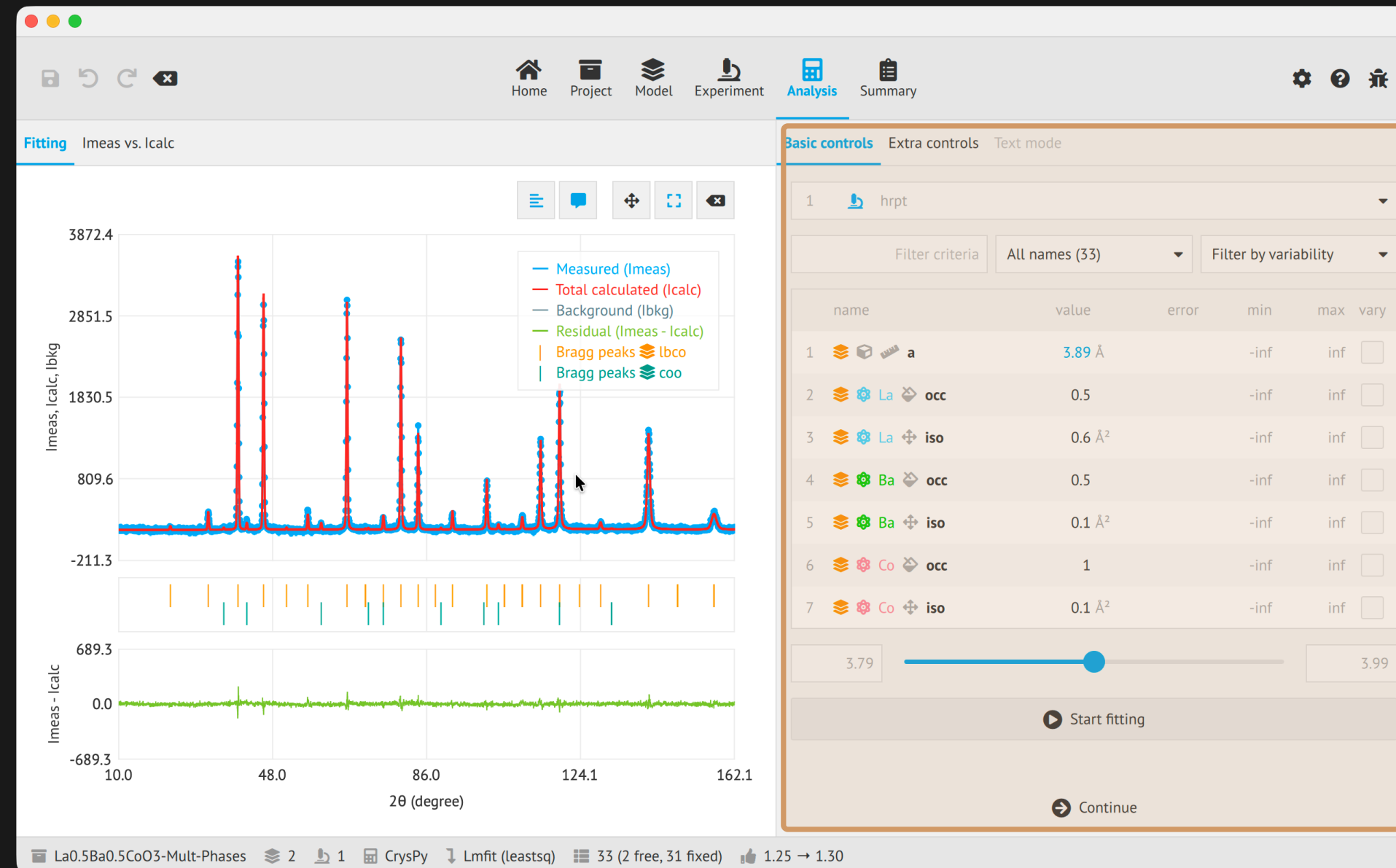
The data — always the largest thing on screen.



One screen, four zones

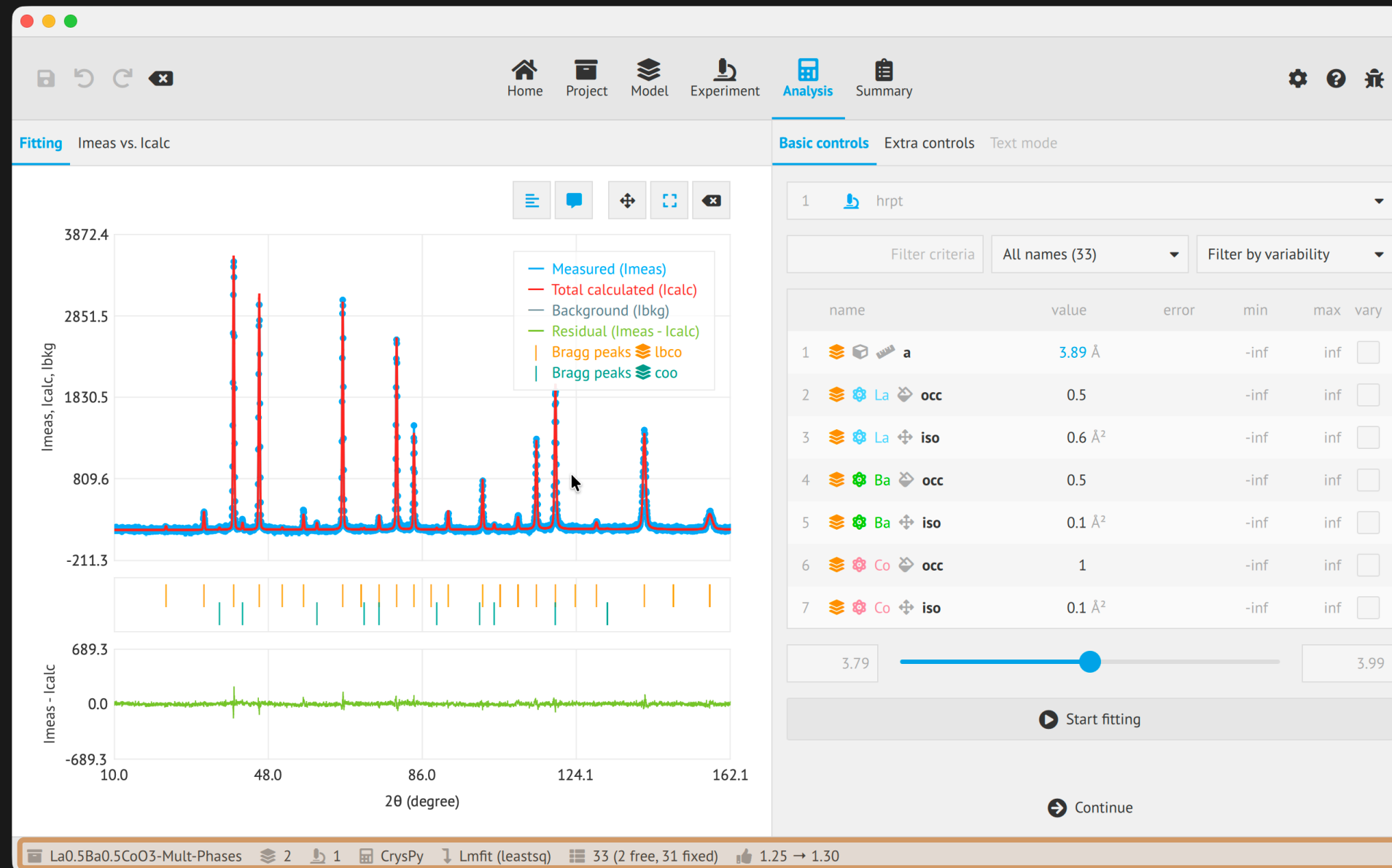
16 | 26

Everything you can change — one column, ending with the fit button.



One screen, four zones

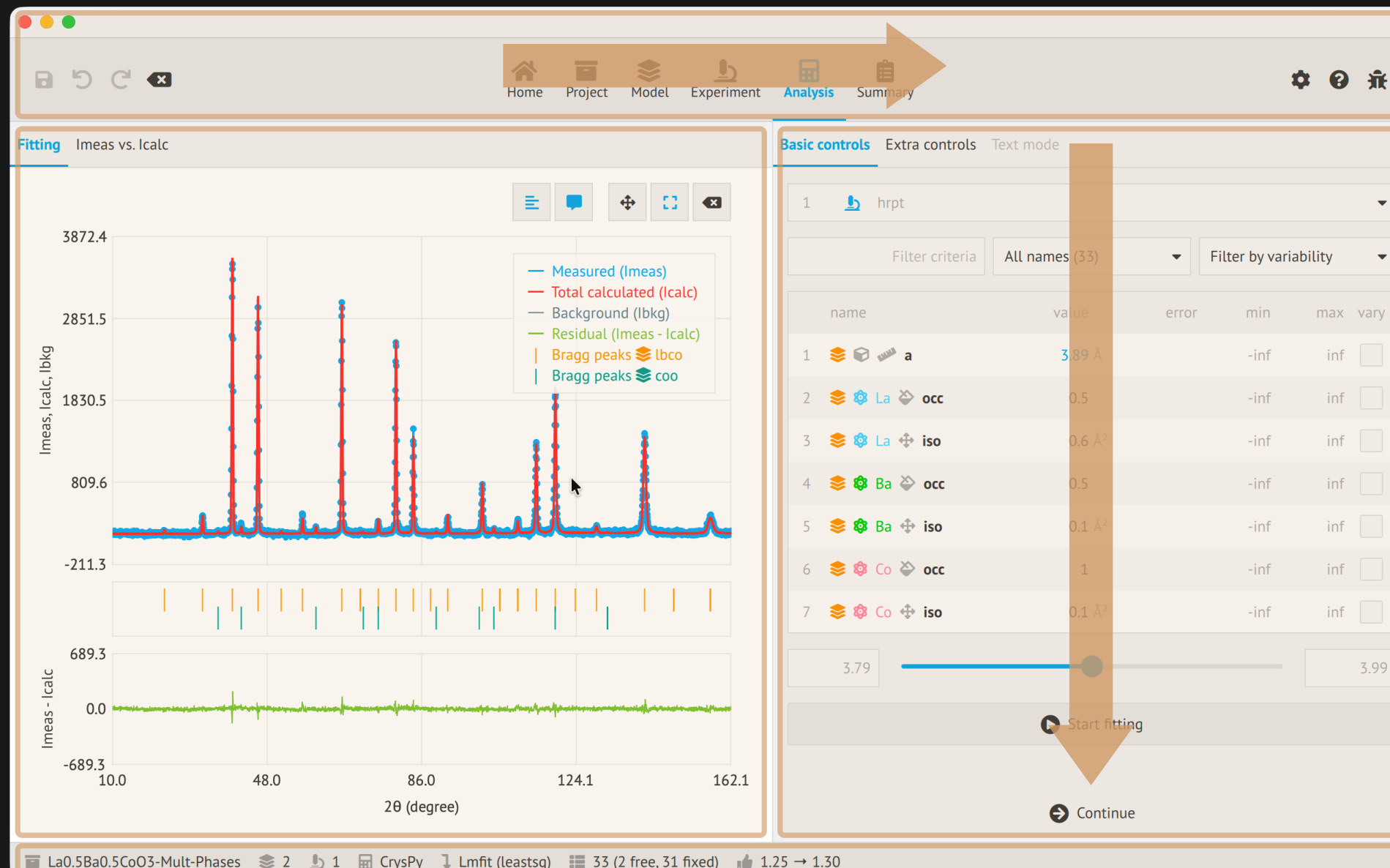
What the program is doing — a line you can ignore, never a dialog.



One screen, four zones

Learn the layout once. It **never moves**.

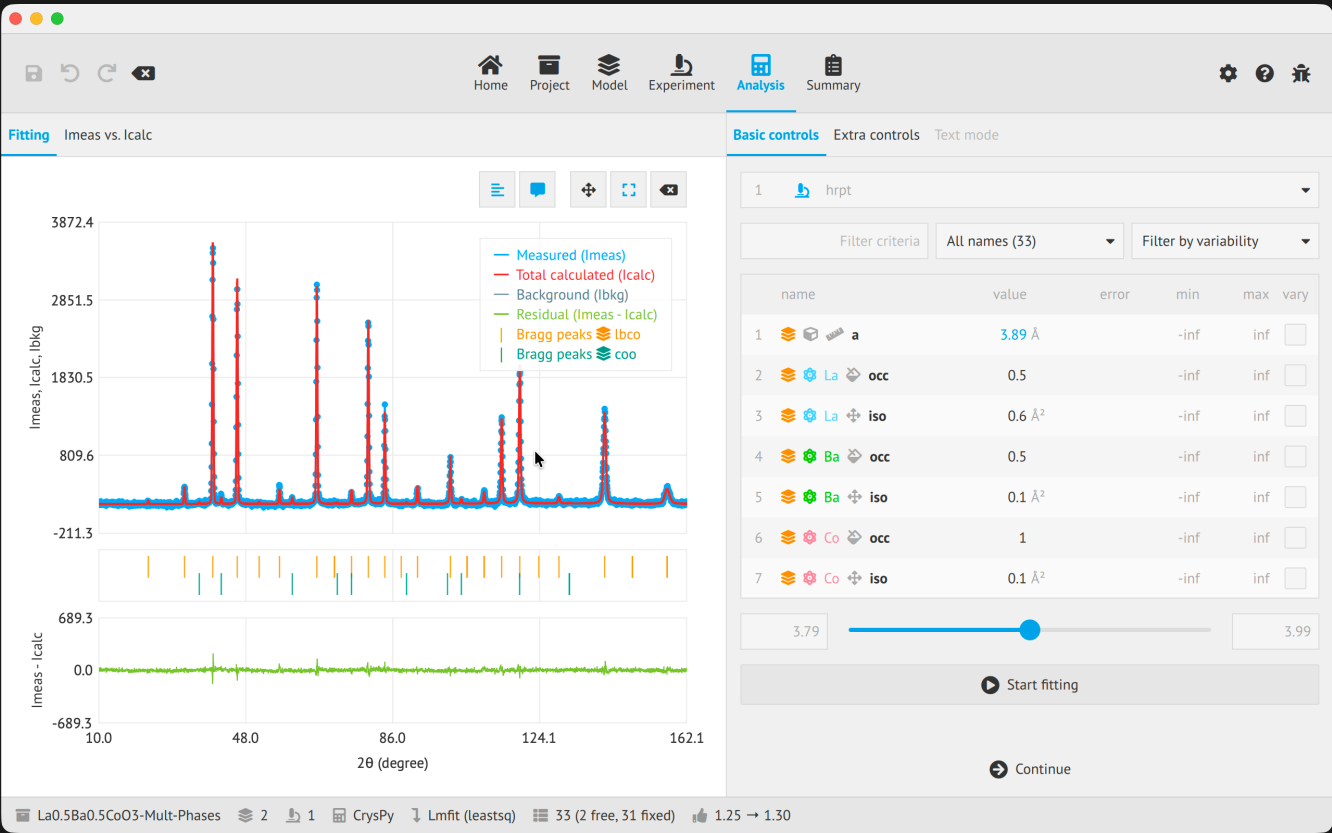
The eye goes **left to right**, then **top to bottom**.



3.4 UX PRINCIPLES

The squint test

Everything on the screen.



What you see in the first second.



The squint test

Only the **structure** survives the blur.

Another interface — nothing stands out.



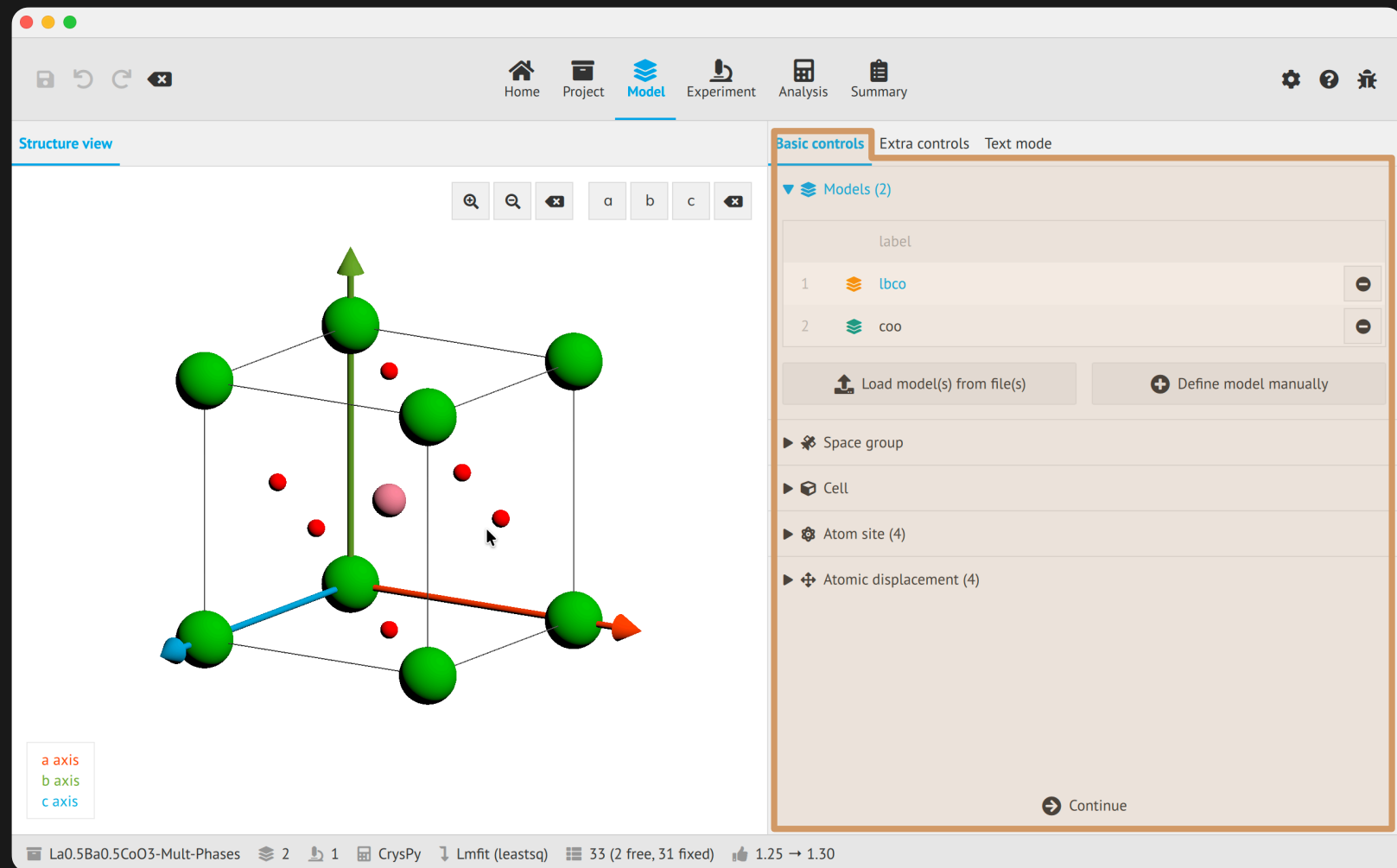
EasyDiffraction — the **four zones** are still there.



Show what is needed, when it is needed

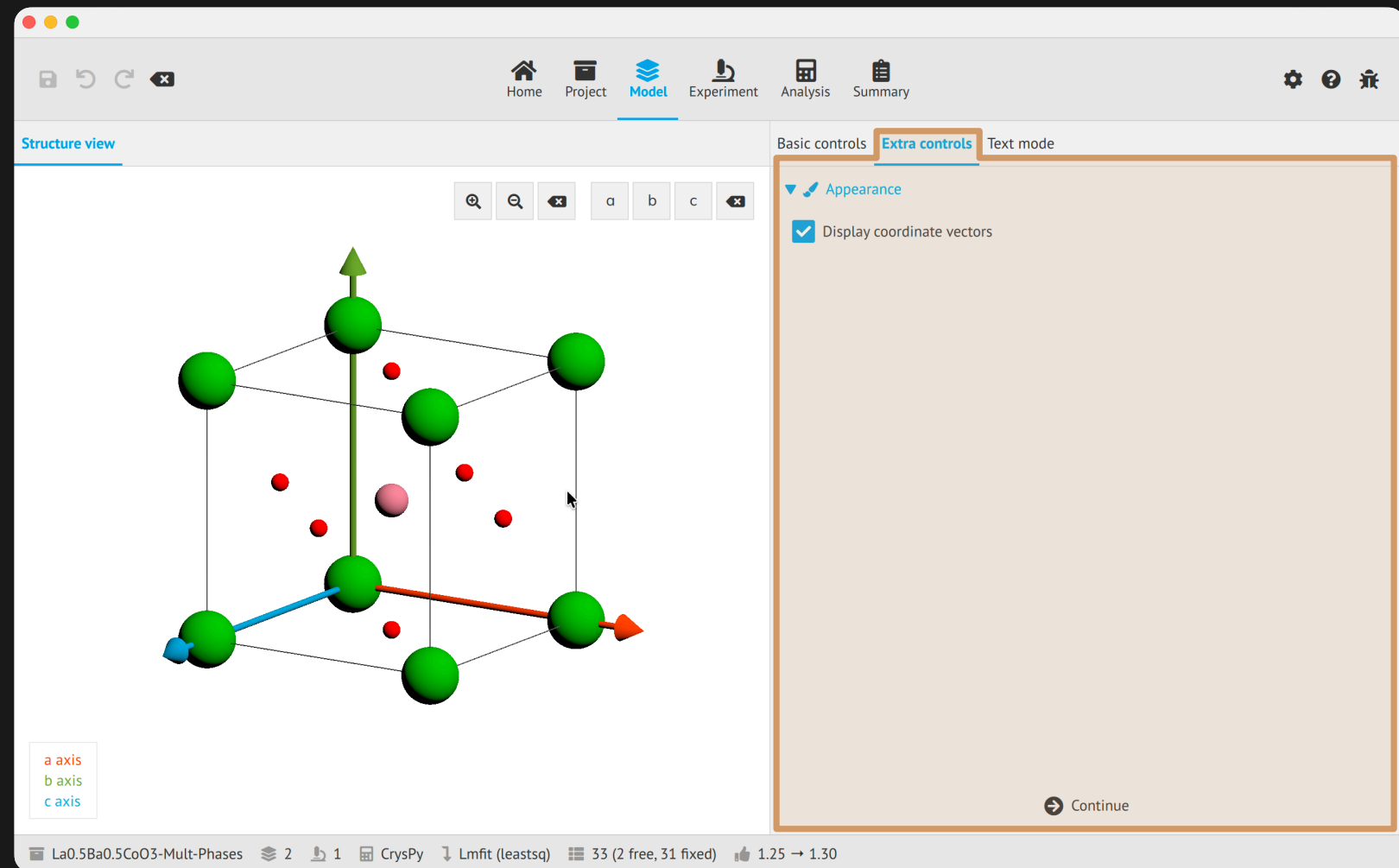
 Basic · Extra · Text mode

three levels, not three screens



Show what is needed, when it is needed

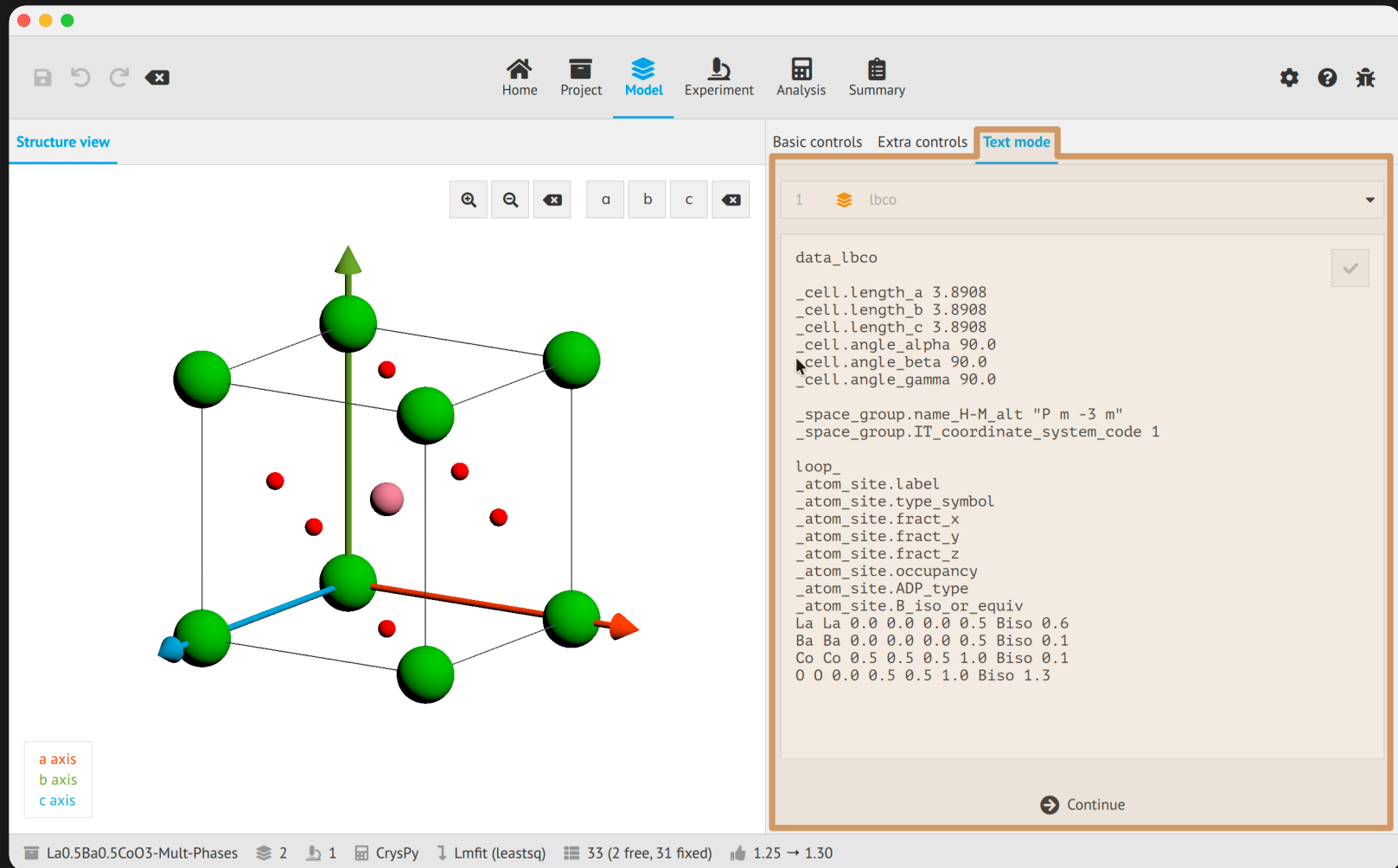
 Basic · **Extra** · Text mode
three levels, not three screens



Show what is needed, when it is needed

 Basic · Extra · **Text mode**

three levels, not three screens



The screenshot displays a software interface for crystallographic analysis. The main window is titled "Structure view" and shows a 3D model of a crystal structure with green spheres representing atoms and red spheres representing smaller atoms. The model is set within a unit cell with axes labeled a, b, and c. A legend in the bottom left corner identifies the axes: a axis (red), b axis (green), and c axis (blue).

On the right side, there is a panel with tabs for "Basic controls", "Extra controls", and "Text mode". The "Text mode" tab is active, showing a text editor with the following content:

```
1 lbco

data_lbco
_cell.length_a 3.8908
_cell.length_b 3.8908
_cell.length_c 3.8908
_cell.angle_alpha 90.0
_cell.angle_beta 90.0
_cell.angle_gamma 90.0

_space_group.name_H-M_alt "P m -3 m"
_space_group.IT_coordinate_system_code 1

loop_
_atom_site.label
_atom_site.type_symbol
_atom_site.fract_x
_atom_site.fract_y
_atom_site.fract_z
_atom_site.occupancy
_atom_site.ADP_type
_atom_site.B_iso_or_equiv
La La 0.0 0.0 0.0 0.5 Biso 0.6
Ba Ba 0.0 0.0 0.0 0.5 Biso 0.1
Co Co 0.5 0.5 0.5 1.0 Biso 0.1
O O 0.0 0.5 0.5 1.0 Biso 1.3
```

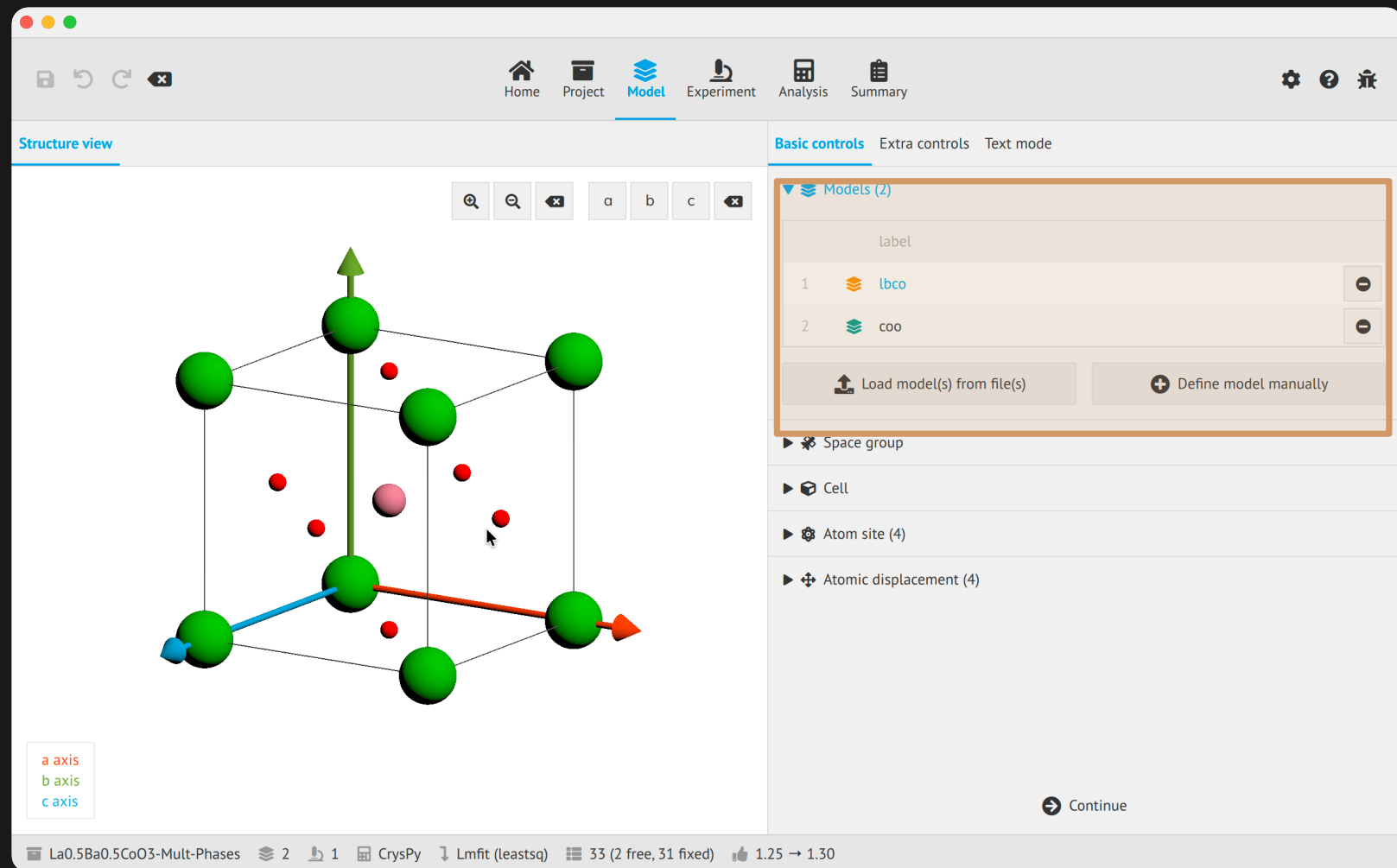
At the bottom of the interface, there is a status bar showing the current project name "La0.5Ba0.5CoO3-Mult-Phases", the number of data points "2", the number of parameters "1", the software used "CrysPy", the refinement method "Lmfit (leastsq)", the number of free and fixed parameters "33 (2 free, 31 fixed)", and the R-factor "1.25 → 1.30".

Show what is needed, when it is needed

 Basic · Extra · Text mode

three levels, not three screens

✓ Sections that collapse
one group open at a time

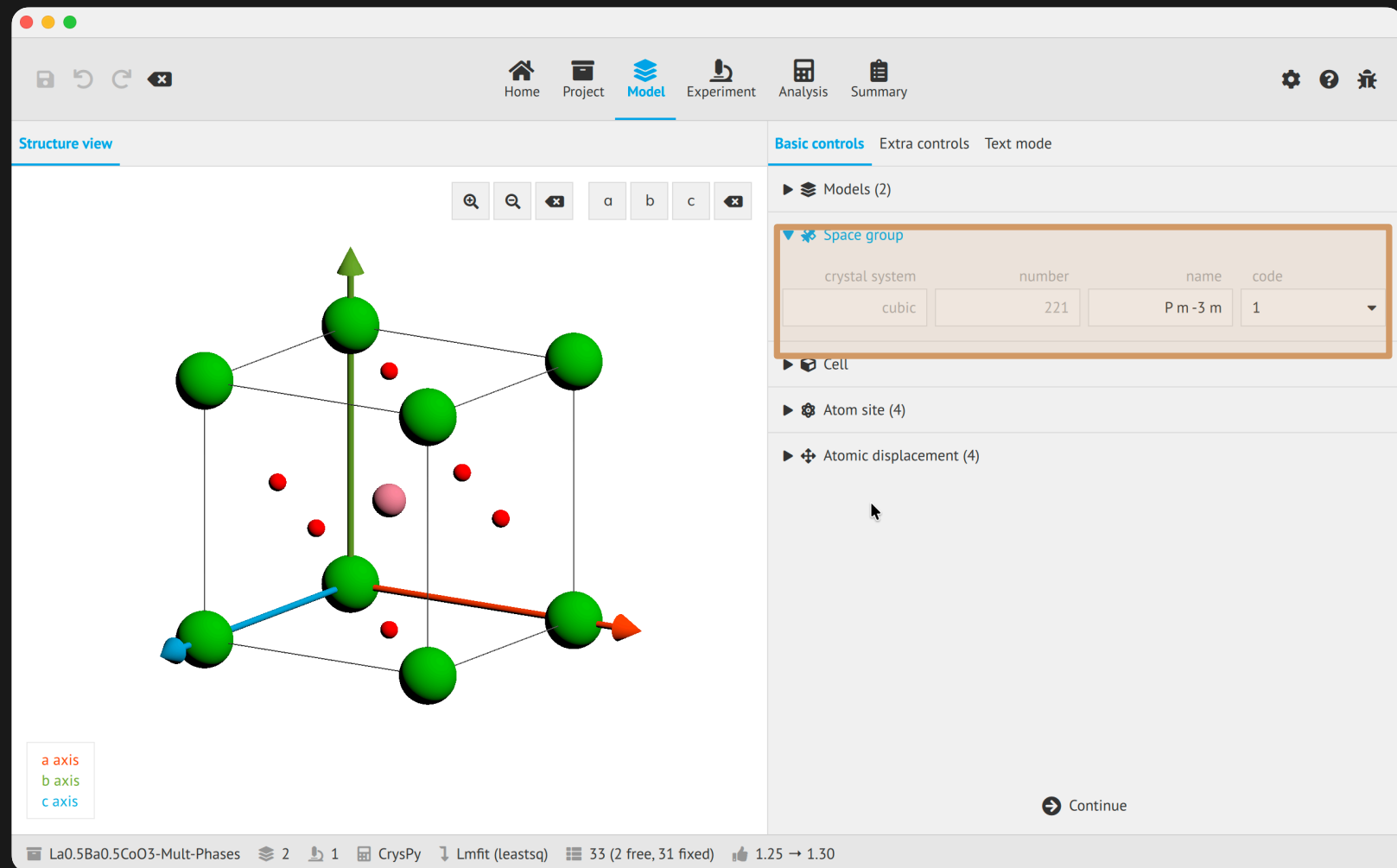


Show what is needed, when it is needed

 Basic · Extra · Text mode

three levels, not three screens

✓ Sections that collapse
one group open at a time

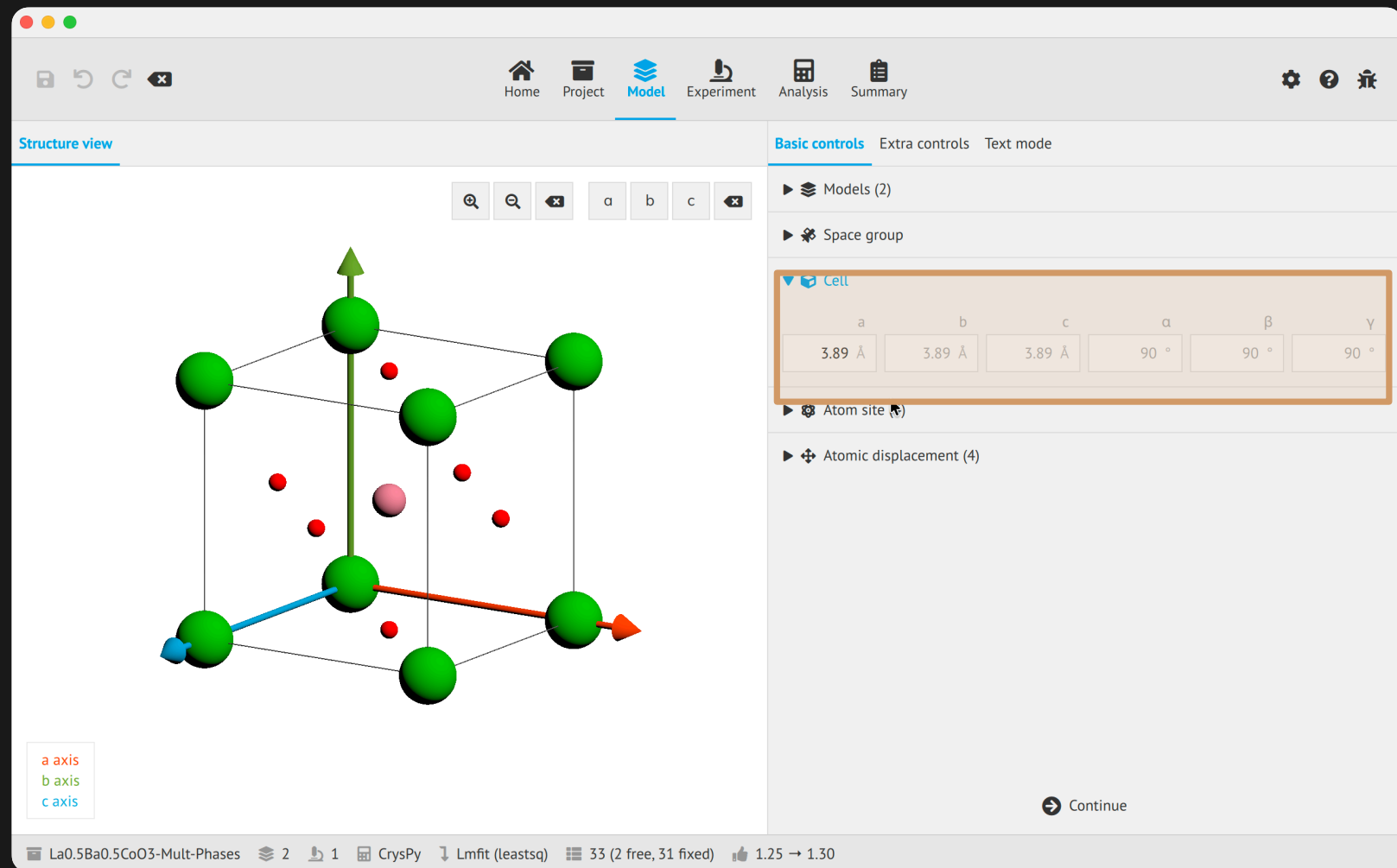


Show what is needed, when it is needed

 Basic · Extra · Text mode

three levels, not three screens

✓ Sections that collapse
one group open at a time



Show what is needed, when it is needed

 Basic · Extra · Text mode

three levels, not three screens

✓ Sections that collapse
one group open at a time

● Greyed, not gone
the screen keeps its shape



Show what is needed, when it is needed

 Basic · Extra · Text mode

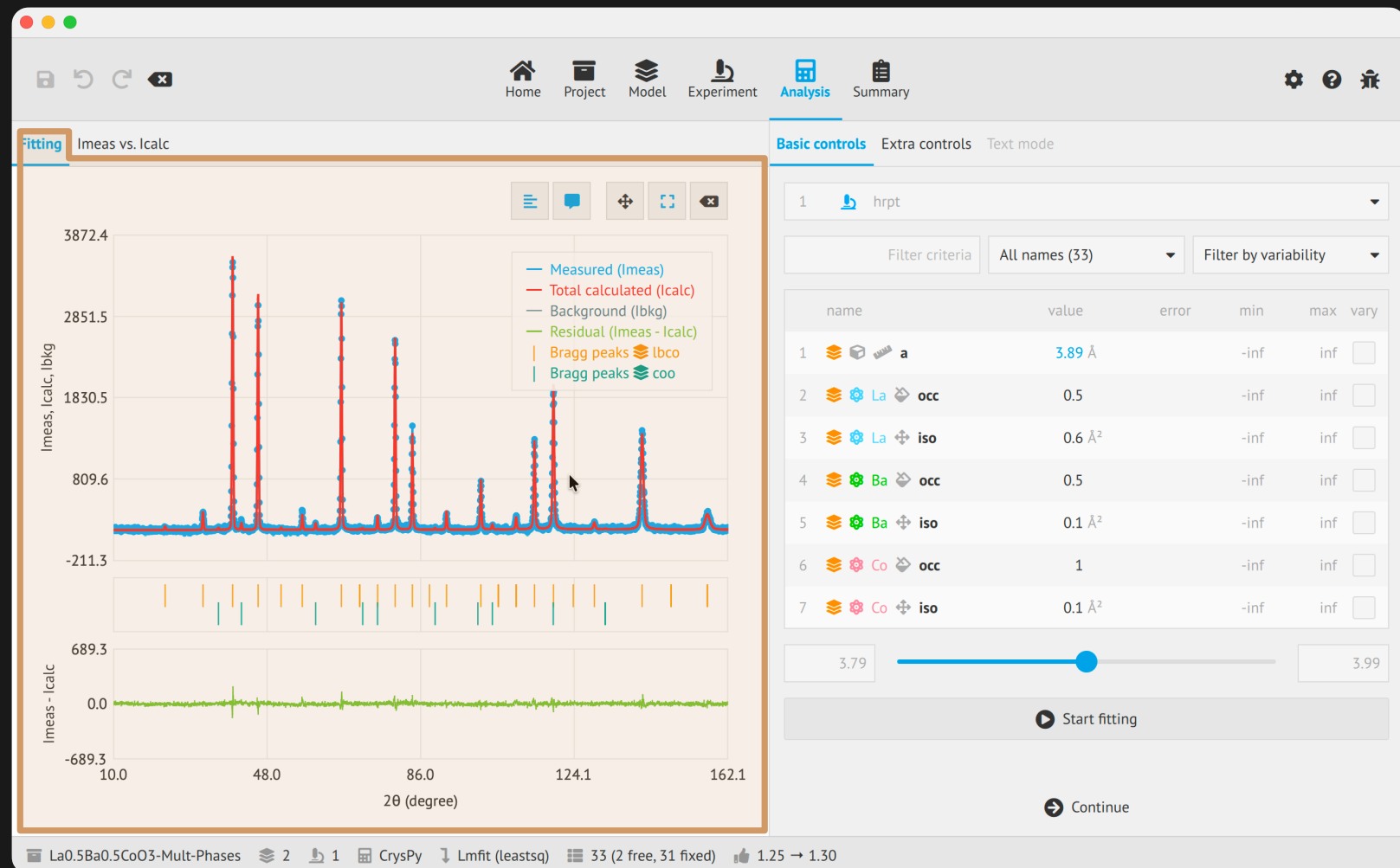
three levels, not three screens

✓ Sections that collapse
one group open at a time

⏸ Greyed, not gone
the screen keeps its shape

 Multiple views

two tabs, not two windows



Show what is needed, when it is needed

Complexity is not removed, only **moved out of the way**.

 **Basic** · Extra · Text mode

three levels, not three screens

✓ Sections that **collapse**

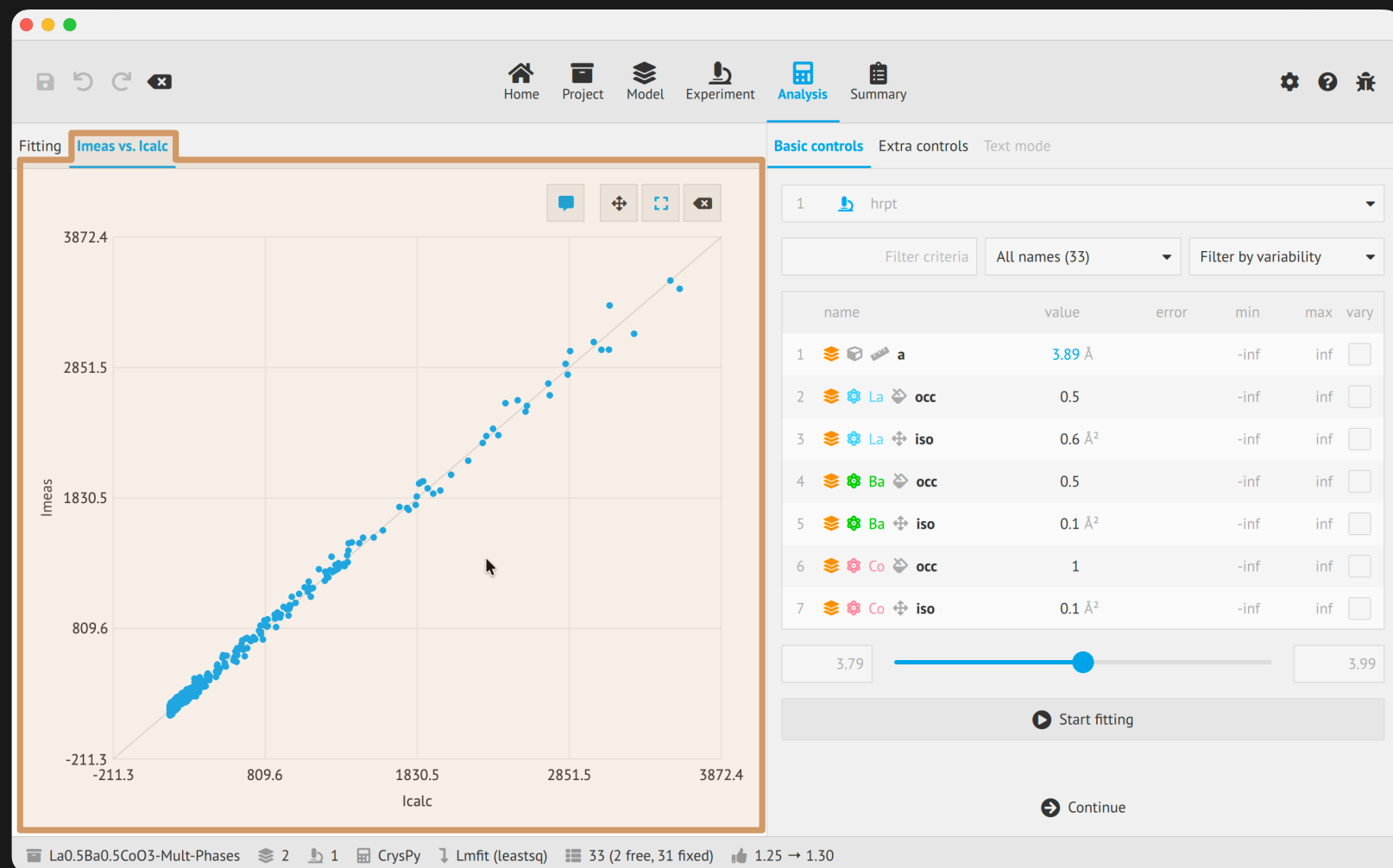
one group open at a time

 Greyed, not **gone**

the screen keeps its shape

 Multiple **views**

two tabs, not two windows



Three ways to run the same workflow

Same objects, same order, in **all three**.

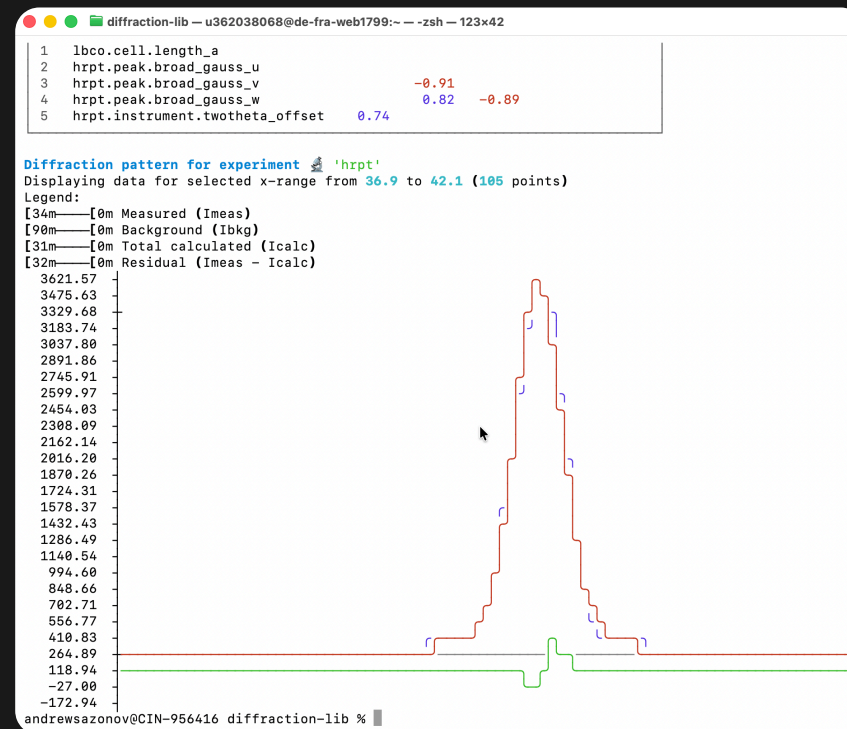
GUI

click through the workflow



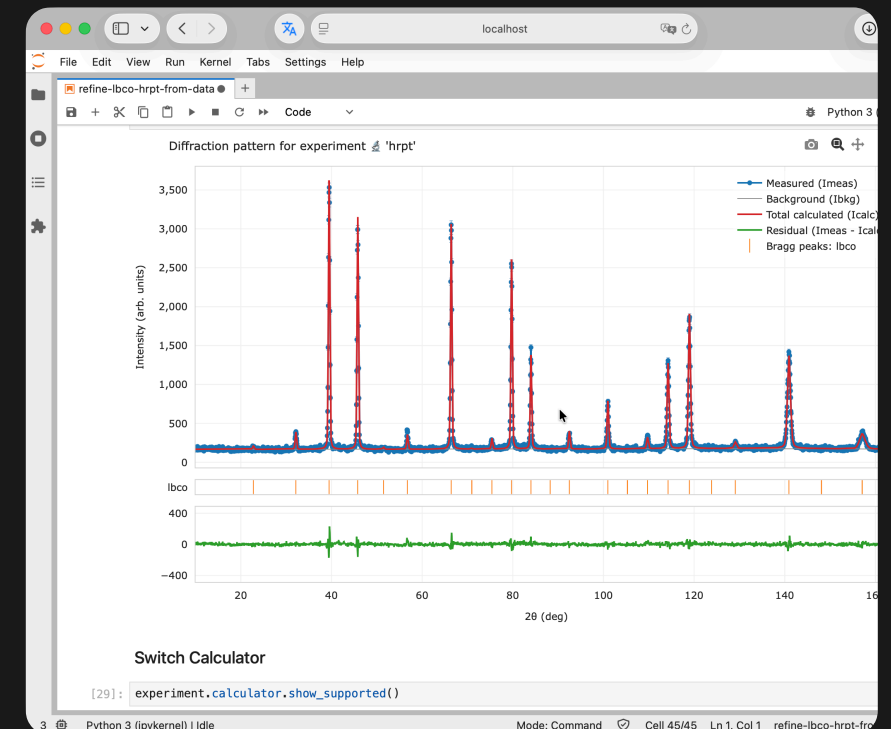
CLI

the same project, one command



Library

the same objects, in Python



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Same screen, different technique

Only the **science** differs.

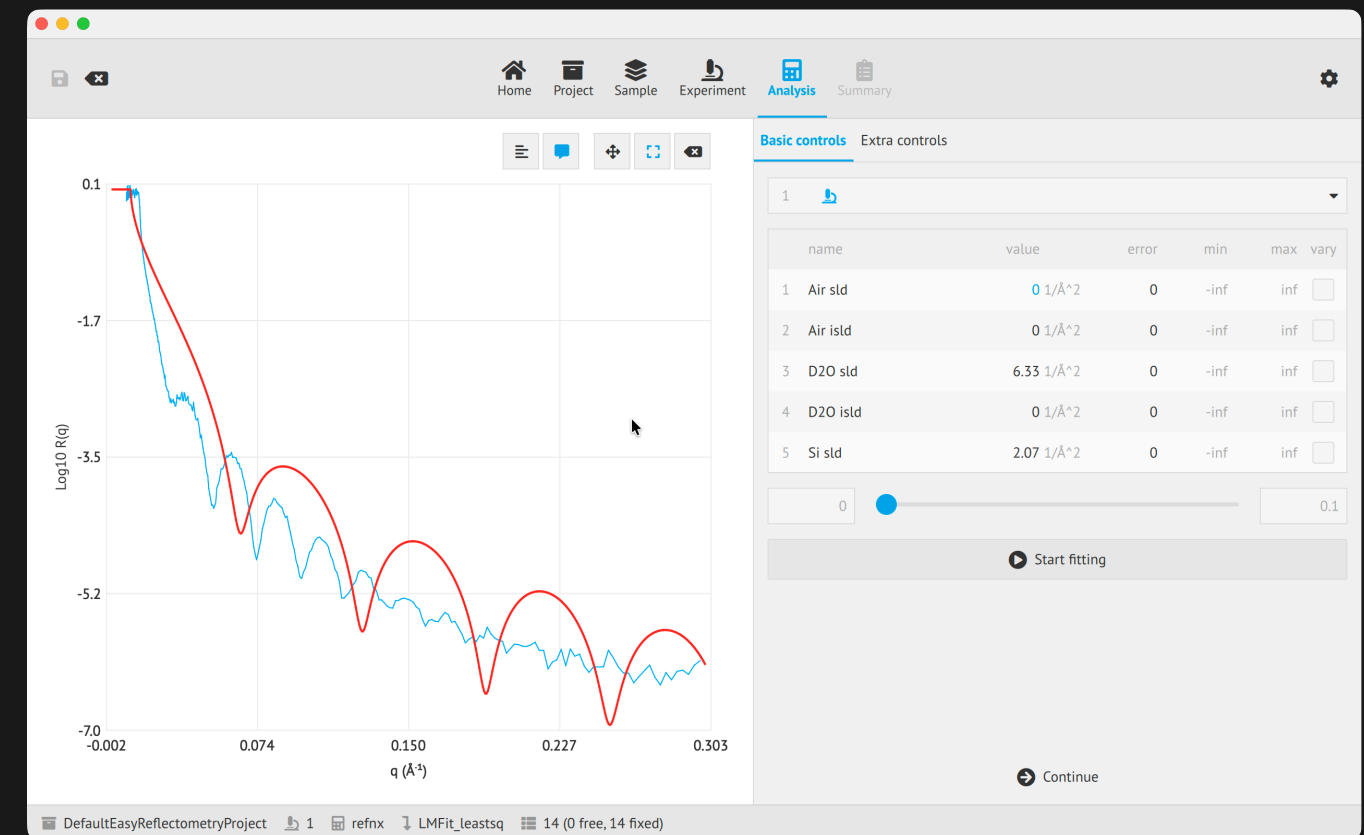
EasyDiffraction

a powder pattern and a crystal structure



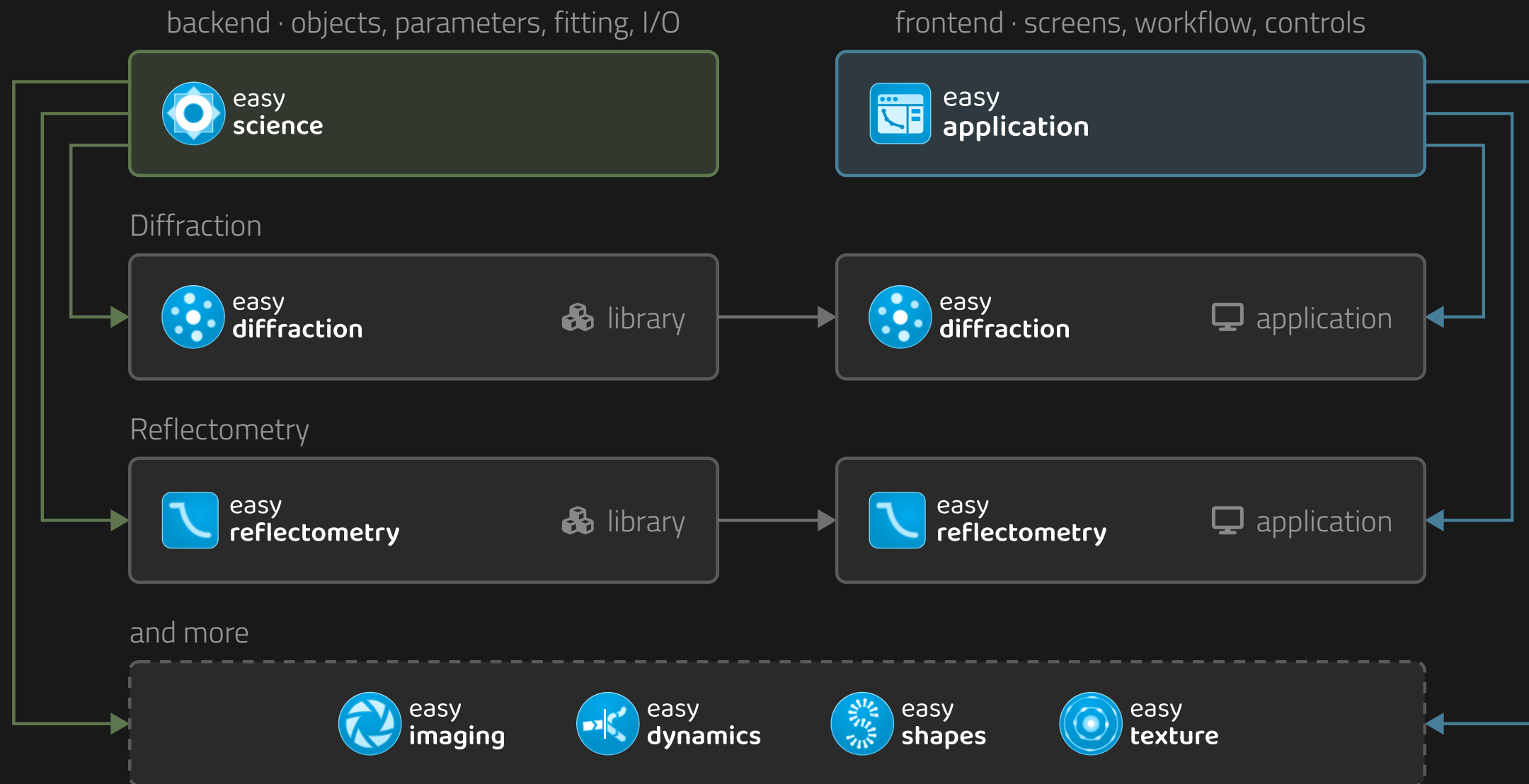
EasyReflectometry

a reflectivity curve and a layer stack



What became reusable

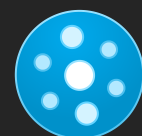
A new technique brings science, **not** a new interface.



Where EasyScience runs today

Different stages, **the same interface**.

easyscience.org · github.com/easyscience



easy
diffraction

diffraction



analysis



running



easy
reflectometry

reflectometry



analysis



running



easy
dynamics

quasielastic neutron scattering



analysis



new



easy
imaging

Bragg-edge imaging



analysis



new



easy
texture

crystallographic texture · built outside ESS



reduction



external



easy
shapes

molecular structure generation · simulation workflows



modelling



new

4.4 REUSE

What reuse brings

Years of design decisions, **inherited for free**.

 Learn one application, and you know how **the others work**

 A usability fix reaches **every** application

 A new technique starts from a **working** interface

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One real refinement



One real refinement

